

Non-Thermal Electromagnetic Field Configurations as Selective Antitumor Monotherapy

Comprehensive cross-paper analysis of 49 in vitro studies

Prepared as a literature-dataset synthesis report

Dataset: FinalTable.xlsx | Analysis date: 5 May 2026

Scope. Rows represent unique experimental conditions. All analyses below use only rows marked as non-thermal after excluding Thermal_Mechanism = 1.

Abstract

This report evaluates whether non-thermal electromagnetic-field (EMF) configurations in the provided 49-paper dataset show evidence of selective antitumor activity in vitro. After excluding 5 thermal rows, 434 non-thermal experimental conditions remained. Rather than accepting study labels at face value, field exposures were re-grouped into seven physics-informed macro-categories: ELF/LF magnetic-field platforms, RF/HF magnetic-field plus static-field hybrids, intermediate-frequency alternating electric fields, nanosecond pulsed electric fields, ultrashort pulsed magnetic fields, RF/mobile-frequency electromagnetic exposures, and millimeter-wave/terahertz radiative exposures. The strongest selectivity signals were not uniformly distributed. Intermediate-frequency TFields/AEF and nsPEF produced the cleanest matched cancer-versus-healthy separation in breast models, especially MDA-MB-231 versus MCF-12 and MCF-7 versus MCF-10A. ELF/LF magnetic-field platforms produced multiple strong effects, including near-complete cancer-cell viability/proliferation loss and striking senescence induction, but the category is heterogeneous and contains both selective and counterproductive windows. MMW/THz radiation produced highly selective apoptosis/senescence in H1299 versus MCF-10A under W-band millimeter-wave exposure, while THz surface irradiation showed less favorable sparing of PBMCs. RF/mobile-frequency and ultrashort pulsed magnetic-field evidence was thinner for monotherapy selectivity because healthy comparators were absent or also affected. The most coherent future hypothesis is that non-thermal EMF selectivity emerges when the exposure couples a physical vulnerability (mitotic spindle alignment, mitochondrial/pore electroporation, oxidative threshold, or membrane/molecular resonance) with a cancer-specific stress state, producing ROS/membrane-potential collapse, mitotic failure, senescence, apoptosis, and/or motility inhibition without a parallel injury signature in matched normal cells.

Report map

- 1. Dataset structure and filtering
- 2. Physics-informed field macro-categories
- 3. Gap identification and evidence coverage
- 4. Subgroup modules: exposure, cell fate, mechanisms, molecular correlates, and limitations
- 5. Cross-subgroup synthesis and experimental hypotheses
- 6. Global limitations and conclusions
- Appendix A. Full exposure-condition lookup tables

1. Dataset structure and non-thermal filtering

Each row is interpreted as one experimental condition rather than one paper. The main analytical blocks are: metadata and cell identity; exposure parameters; cell-fate outcomes; mechanistic outcomes; and genomic/proteomic molecular outcomes. Direction variables are used as significance-aware signs, while %_vs_Control values are used to rank effect magnitude. For mechanisms that are flags or qualitative mentions, the report distinguishes directly measured effects from mechanisms that were only mentioned.

Table 1. Thermal filtering and retained analysis base.

Metric	Value
Total rows in workbook	439
Rows excluded because Thermal_Mechanism = 1	5 (P3:2; P35:3)
Rows retained for non-thermal analysis	434
Distinct papers retained	49
Cancer rows retained	339
Healthy/non-cancer rows retained	95

2. Physics-informed macro-categories

Table 2. Emergent macro-categories from the observed Field_Type labels.

Macro-category	Original Field_Type labels merged	Justification
ELF/LF magnetic-field platforms	ELF-EMF, ELF-MF (ELF-EMF), EMF (LF-EMF), AMF (LF-EMF), MEMF (ELF-EMF), ELF-EMF + SMF, SMF, PEMF (ELF-EMF), ELF P-EMF (ELF-EMF), ELF-PEMF (ELF-EMF), SREMF (ELF-EMF), Spinning Oscillating Magnetic Field (ELF-MF), Gradient RMF (ELF-MF), LF-RMF (ELF-RMF), Oscillating Magnetic Field - Helmholtz coil, fixed axis (ELF-EMF)	Merged because the common dominant exposure is quasi-static/low-frequency magnetic induction, including continuous sinusoidal ELF fields, rotating magnetic fields, PEMF/MEMP-like pulsed magnetic fields, low-frequency alternating magnetic fields, and static-field-biased ELF conditions. Although this category is heterogeneous, the shared physics is magnetic-field dominance with negligible direct electric-field reporting and frequencies from sub-Hz to low hundreds of kHz.
RF/HF magnetic field + static-field hybrids	RF-MF + SMF (MF/HF-EMF + SMF)	Kept separate because it combines MHz magnetic components with defined static fields and explicitly targets weak-field mitochondrial hyperfine/resonance mechanisms. Its AC amplitudes are nT-uT and therefore physically distinct from the mT-T ELF/LF magnetic platforms.
Intermediate-frequency alternating electric fields	TTFields (AEF); Tumor Treating Induced Fields (TTIF); AEF	Merged because the central exposure is an alternating electric field intended to perturb proliferating cells, mostly in the 100-300 kHz range with V/cm-level fields. The standalone AEF row is retained here because it is electric-field dominant and biologically framed as cancer-selective, although it is not a canonical TTFields device.
Nanosecond pulsed electric fields	nsPEF	Kept separate because the active physical event is high-field, nanosecond-duration electroporation/electroperturbation. Frequency is not the organizing variable; pulse width, kV/cm amplitude, repetition rate, and pulse number define the dose.
Ultrashort pulsed magnetic fields	nsPMF; μ sPMF	Merged because both are pulsed magnetic fields with ns-us pulse widths and pulse/dB/dt-driven dose logic. The category is separated from ELF/LF PEMF because the reported intensities and pulse kinetics are much faster and the available evidence is a single C6 glioma paper.
RF/mobile-frequency electromagnetic exposures	LTE (UHF-EMF); RF-EMF (MF-EMF); REAC (RF-EMF)	Merged because these are RF-band electromagnetic exposures below millimeter wavelengths, including mobile-communication SAR-based exposures, 13.56 MHz RF, and radioelectric conveyance. The dominant dose metrics are SAR, device power, or weak RF stimulation rather than magnetic flux density.

Macro-category	Original Field_Type labels merged	Justification
Millimeter-wave/terahertz radiative exposures	MMW (EHF-EMF); THz radiation (EHF-EMF)	Merged because both are high-frequency radiative exposures where power density and frequency band dominate, with non-thermal mechanisms proposed around biomolecular/membrane modes. This category is distinct from RF/mobile because penetration, dosimetry, and resonance rationales differ strongly.

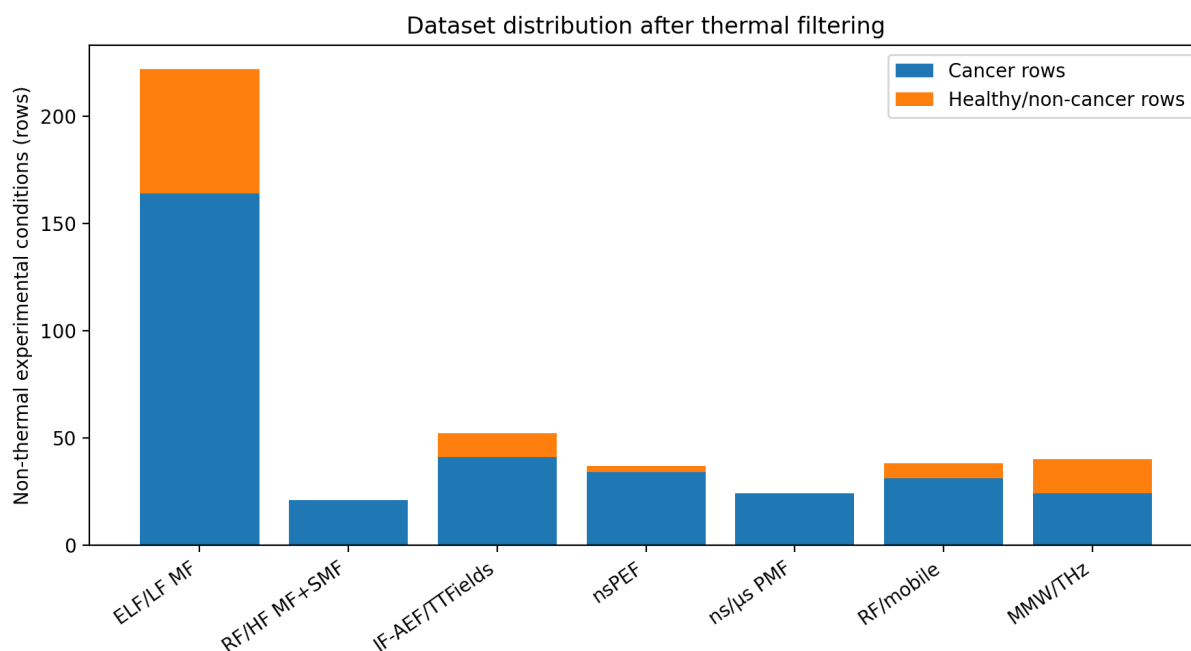


Figure 1. Non-thermal row distribution by macro-category and cell status.

3. Gap identification and evidence coverage

Coverage is reported as experimental-condition coverage and independent-paper coverage. A high row percentage driven by one paper is treated as weaker evidence than a lower percentage reproduced across multiple Paper_IDs. The strongest evidence bases are not simply the largest row counts: for example, ultrashort pulsed magnetic fields have 100% viability and morphology coverage, but all 24 rows come from one paper and one cancer cell line.

Table 3. Gap identification by field macro-category.

Macro-category	Evidence base	Cancer/healthy rows	Measured coverage highlights (* = ≥ 10 rows from one paper)	Unmeasured blocks
ELF/LF magnetic-field platforms	222 rows; 27 papers	164C/58H	Viability 157/222 (20P); Proliferation 116/222 (20P); Apoptosis 52/222 (12P); Necrosis 26/222 (7P); Ferroptosis 2/222 (1P); Senescence 3/222 (1P); Motility 17/222 (6P); ROS 36/222 (10P); ...	
RF/HF magnetic field + static-field hybrids	21 rows; 2 papers	21C/0H	Viability 7/21 (1P); Proliferation 3/21 (2P); ROS 16/21 (2P); Antioxidant 7/21 (1P); Membrane Potential 10/21 (1P)*; Calcium 8/21 (1P)	Apoptosis; Necrosis; Ferroptosis; Senescence; Motility; Cellular Morphology; Cytoskeleton; Metabolic Reprogramming; ...
Intermediate-frequency alternating electric fields	52 rows; 6 papers	41C/11H	Viability 35/52 (5P); Proliferation 19/52 (2P); Apoptosis 5/52 (2P); Motility 8/52 (3P); ROS 1/52 (1P); Membrane Potential 1/52 (1P);	Necrosis; Ferroptosis; Senescence; Antioxidant; Metabolic Reprogramming; Differentiation; Cell Cycle

Macro-category	Evidence base	Cancer/healthy rows	Measured coverage highlights (* = >=10 rows from one paper)	Unmeasured blocks
			Calcium 1/52 (1P); Cellular Morphology 19/52 (5P); ...	
Nanosecond pulsed electric fields	37 rows; 5 papers	34C/3H	Viability 15/37 (4P); Proliferation 5/37 (3P); Apoptosis 5/37 (2P); Motility 2/37 (1P); ROS 2/37 (1P); Membrane Potential 22/37 (2P); Calcium 2/37 (1P); Cellular Morphology 29/37 (5P); ...	Necrosis; Ferroptosis; Senescence; Antioxidant; Differentiation; Cell Cycle
Ultrashort pulsed magnetic fields	24 rows; 1 papers	24C/0H	Viability 24/24 (1P)*; Cellular Morphology 24/24 (1P)*	Proliferation; Apoptosis; Necrosis; Ferroptosis; Senescence; Motility; ROS; Antioxidant; ...
RF/mobile-frequency electromagnetic exposures	38 rows; 4 papers	31C/7H	Viability 16/38 (2P); Proliferation 16/38 (2P); Apoptosis 13/38 (3P); Senescence 7/38 (2P); ROS 4/38 (1P); Cellular Morphology 3/38 (1P); Cell Cycle 7/38 (2P)	Necrosis; Ferroptosis; Motility; Antioxidant; Membrane Potential; Calcium; Cytoskeleton; Metabolic Reprogramming; ...
Millimeter-wave/terahertz radiative exposures	40 rows; 4 papers	24C/16H	Viability 12/40 (2P); Apoptosis 26/40 (2P); Necrosis 4/40 (1P); Senescence 22/40 (1P)*; Cellular Morphology 29/40 (3P)	Proliferation; Ferroptosis; Motility; ROS; Antioxidant; Membrane Potential; Calcium; Cytoskeleton; ...

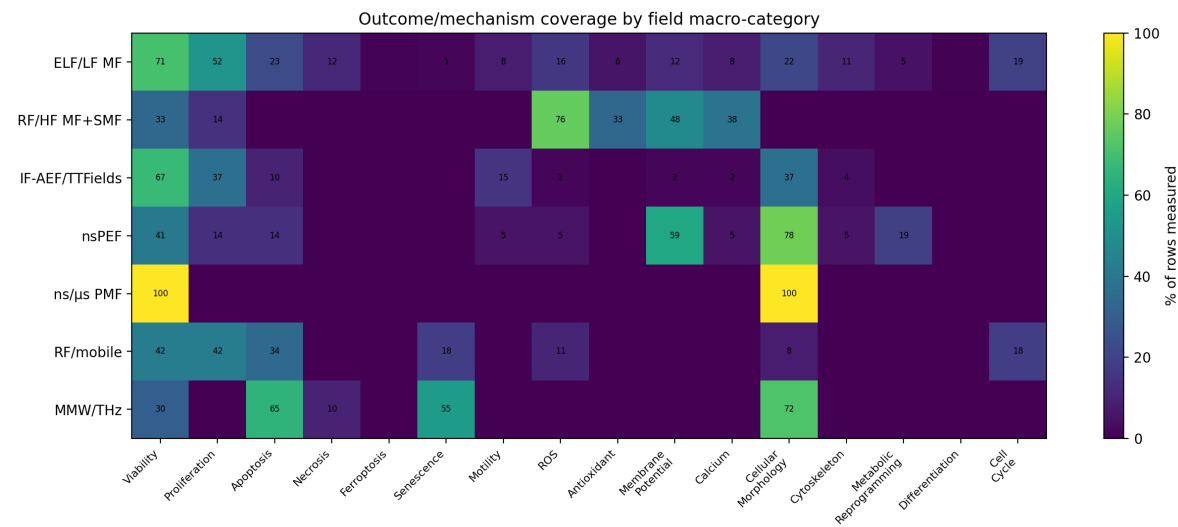


Figure 2. Coverage heatmap; cell values are percent of rows in the macro-category in which the endpoint/mechanism was measured.

Major gaps are immediately visible. Ferroptosis and differentiation are almost absent. Antioxidant response is sparse outside ELF/LF and RF/HF magnetic-field hybrids. Calcium is directly measured in few conditions and otherwise often only mentioned. Molecular profiling is uneven: ELF/LF fields have the broadest gene/protein coverage, while several categories have no genomic/proteomic data at all. Healthy comparators are particularly underrepresented for RF/HF magnetic hybrids, ultrashort pulsed magnetic fields, and several cancer-effective nsPEF conditions.

4. Subgroup-level analysis

4.1. ELF/LF magnetic-field platforms

4.1.1 Subgroup overview

This is the largest and most heterogeneous subgroup. It spans power-line frequencies, low-frequency rotating/gradient fields, pulsed ELF devices, static-field-biased ELF exposure, and 100-320 kHz alternating magnetic fields. It contains the broadest cell-line landscape and the largest number of matched cancer/healthy experiments, but strong signals are

distributed across distinct physical subfamilies rather than one uniform dose-response axis. The retained evidence base is 222 conditions from 27 papers, with a cancer-to-healthy row ratio of 164:58.

Per-study cancer/healthy ratio: P1:17C/9H; P2:17C/0H; P3:2C/0H; P4:17C/3H; P5:7C/11H; P6:4C/1H; P8:8C/0H; P9:2C/1H; P10:2C/1H; P13:3C/1H; P14:14C/3H; P15:1C/0H; P16:10C/2H; P18:7C/2H; P19:12C/6H; P20:2C/2H; P21:3C/2H; P23:6C/0H; P24:3C/1H; P25:1C/0H; P28:3C/1H; P36:2C/0H; P38:8C/8H; P40:4C/2H; P41:1C/1H; P44:4C/0H; P48:4C/1H.

Direct selectivity is possible in papers with both cancer and healthy/non-cancer lines: P1, P4, P5, P6, P9, P10, P13, P14, P16, P18, P19, P20, P21, P24, P28, P38, P40, P41, P48.

Dominant tested biological contexts: Glioblastoma (31 rows); Triple-negative breast cancer (TNBC) (27 rows); Breast cancer (20 rows); Neuroblastoma (13 rows); Astrocytes (9 rows).

Table 4.1a. Cell-line panel for ELF/LF magnetic-field platforms.

Role	Tumour/tissue type	Species/origin	Cell lines (number of conditions)	Rows
Cancer	Triple-negative breast cancer (TNBC)	Human	MDA-MB-231 (25), BT549 (1), MDA-MB-468 (1)	27
Cancer	Glioblastoma	Human	U87 (6), LN229 (5), 1627 (4), GBM115 (4), U251 (2), T98 (1), A172 (1)	23
Cancer	Breast cancer	Human	MCF-7 (14), SKBR3 (5), MDA-MB-453 (1)	20
Cancer	Neuroblastoma	Mouse	N2A (9)	9
Cancer	Pancreatic cancer	Human	PANC-1 (9)	9
Cancer	Glioblastoma	Mouse	CT2A (8)	8
Cancer	Fibrosarcoma	Human	HT-1080 (7)	7
Cancer	Human lymphoma cells (Burkitt lymphoma)	Human	Raji (6)	6
Cancer	Melanoma	Mouse	B16F10 (3), B16-BL6 (3)	6
Cancer	Pheochromocytoma	Rat	PC12 (5)	5
Cancer	Cervical adenocarcinoma	Human	HeLa (4)	4
Cancer	Cervical cancer	Human	HeLa-LUC (1), CaSki-LUC (1), SiHa-LUC (1), HeLa-LUC (co-culture with CD8+ T cells) (1)	4
Cancer	Diffuse Intrinsic Pontine Glioma	Human	DIPG (4)	4
Cancer	Glioblastoma multiforme	Human	T98G (4)	4
Cancer	Neuroblastoma	Human	SH-SY5Y (4)	4
Cancer	Glioblastoma astrocytoma	Human	U373 MG (3)	3
Cancer	Prostate cancer	Human	DU-145 (3)	3
Cancer	Clear cell renal cell carcinoma (ccRCC)	Human	786-O (1), 769-P (1)	2
Cancer	Colon adenocarcinoma	Mouse (C57BL/6)	MC-38-Luc (2)	2
Cancer	Human liver cancer cells (hepatocellular carcinoma)	Human	HepG2 (2)	2
Cancer	Human non-small-cell lung cancer cells	Human	A549 (2)	2
Cancer	Lung adenocarcinoma	Human	A549 (2)	2
Cancer	Bladder cancer (urothelial cell carcinoma)	Human	HT-1197 (1)	1
Cancer	Breast duct carcinoma (ER+)	Mouse	MC4-L2 (1)	1
Cancer	Chronic myelogenous leukemia	Human	K562 (1)	1
Cancer	Colorectal carcinoma (primary tumor)	Human	HT-29 (1)	1
Cancer	Glioblastoma (P53-overexpressing transfected variant of U251)	Human	U251-P53 (1)	1
Cancer	Lung carcinoma	Mouse	LLC (1)	1
Cancer	Metastatic clear cell renal cell carcinoma (ccRCC)	Human	Caki1 (1)	1
Cancer	Nephroblastoma (Wilms tumor)	Human	G401 (1)	1
Healthy/non-cancer	Astrocytes	Mouse	C8D1A (9)	9
Healthy/non-cancer	Breast epithelial cells	Human	MCF-10A (8)	8
Healthy/non-cancer	Normal astrocytes	Human	NHA (5)	5
Healthy/non-cancer	Dermal fibroblasts	Human	HDFa (4)	4
Healthy/non-cancer	Fibroblast	Mouse	3T3 (3)	3
Healthy/non-cancer	Mouse fibroblasts	Mouse	A9 (3)	3
Healthy/non-cancer	Alveolar macrophages (primary cells)	Porcine	PAM (2)	2
Healthy/non-cancer	Embryonic kidney cells	Human	HEK293T (2)	2
Healthy/non-cancer	Embryonic kidney epithelial	Human	HEK293 (2)	2
Healthy/non-cancer	Fibroblast	Human	HDF (Human Dermal Fibroblast) (2)	2
Healthy/non-cancer	Keratinocyte (skin)	Human	HaCaT (2)	2

Role	Tumour/tissue type	Species/origin	Cell lines (number of conditions)	Rows
Healthy/non-cancer	Kidney fibroblasts	African green monkey	CV-1 (2)	2
Healthy/non-cancer	Normal fibroblast	Mouse	NIH/3T3 (2)	2
Healthy/non-cancer	Normal human renal epithelial cells	Human	293T (2)	2
Healthy/non-cancer	SV40-transformed kidney cells	African green monkey	COS-1 (2)	2
Healthy/non-cancer	Umbilical vein endothelial cells	Human	HUVEC (2)	2
Healthy/non-cancer	Cardiac fibroblast	Human	HCF (1)	1
Healthy/non-cancer	Cytotoxic T lymphocytes	Human	CD8+ T cells (from CC patient tumor tissues) (1)	1
Healthy/non-cancer	Lung lymphatic microvascular endothelial cells	Human	hLEC (1)	1
Healthy/non-cancer	Normal dermal fibroblast	Human	FF95 (1)	1
Healthy/non-cancer	Normal fibroblast	Human	Normal human fibroblast (1)	1
Healthy/non-cancer	Retinal pigment epithelial	Human	RPE1 (1)	1

4.1.2 Exposure parameters and rationale

The subgroup contains 67 distinct physical exposure signatures after collapsing rows with identical field parameters. Full condition IDs are provided in Appendix A and are used throughout this module.

Table 4.1b. Parameter envelope for ELF/LF magnetic-field platforms.

Parameter	Values present in retained non-thermal rows
Field labels	ELF-EMF; ELF P-EMF (ELF-EMF); AMF (LF-EMF); MEMP (ELF-EMF); Spinning Oscillating Magnetic Field (ELF-MF); Oscillating Magnetic Field - Helmholtz coil, fixed axis (ELF-EMF); ELF-EMF + SMF; ELF-PEMF (ELF-EMF); SMF; ELF-MF (ELF-EMF); SREMF (ELF-EMF); Gradient RMF (ELF-MF); EMF (LF-EMF); LF-RMF (ELF-RMF); PEMF (ELF-EMF)
Waveforms	Square pulse; Sine; Pulsed (multifrequency); Full square (pulsed); Square; Sine-like (not explicitly specified); Complex patterned (modeled after intercellular calcium wave firing pattern); Periodic
AC frequencies	numeric envelope 0.01-320000; 20 Hz; 30 Hz; 45 Hz; 50 Hz; 55 Hz; 60 Hz; 40 Hz; 3 Hz; 6 Hz; 10 Hz; 14 Hz; 4 Hz; 8 Hz; 227000 Hz; 198000 Hz; 160000 Hz; +22 more
AC magnetic intensity	numeric envelope -16-600; 100 μ T; 70 μ T; 22.6 mT; 35 mT; 18 mT; >2 T; 100 mT; 5 mT; 1.0-2.5 mT; 0.011 T; 1.5-16 mT; 10 μ T; 18 μ T; 18.5 μ T; 0.1 mT; 1 mT; +15 more
DC/static field	numeric envelope -45-45; 45 μ T; 3.0-4.0 mT; -45 μ T; 15 mT
Devices	RILZ coil system (custom design, CTB – Universidad Politécnica de Madrid); Coaxial coils; Solenoid coil (6000 turns); Transistor inverter (Hot Shot) + solenoid copper coil; Modulator equipment by Paso Alto Biotechnology Inc.; ELF-EMF exposure device (Parto Farazane Farda, Iran) connected to a GWInstek SFG-1000 series waveform generator; Oncomagnetic device (oncoscillator with spinning neodymium N52 permanent magnet). 1Dsp.; Helmholtz coil driven by function generator (Wavetek) and high current amplifier (Taidacent); Self-constructed MF exposure facility based on a prototype by Tofani et al., consisting of a pair of horizontally mounted coils connected with a circuit providing DC and 50 Hz AC.; Alphatron 4100-MW; Custom-built ELF-EMF generator with iron core magnetic circuit (two 500-turn coils in series, 1000 turns total), capacitor bank for LC resonance, housed in a Teflon-sealed box placed inside a CO2 incubator. Cooling unit with 5°C circulating water maintained 37°C in the exposure volume.; RILZ coils (custom-designed capsule-shaped coil system with two coils of 222 turns of enameled copper each, powered by a microprocessor-controlled current source); Helmholtz coil (11-inch diameter, 500 turns of 26-gauge copper wire) connected to a frequency generator, verified by an oscilloscope; Self-made magnetic field generator (enameled copper wire in ABS plastic shell, 450x230x25 mm); Square Helmholtz coil inside Mu-metal shielded box, driven by FY6900 DDS waveform generator (Kkmoon) and HP 33120A 15 MHz waveform generator; HP E3631A power supply for DC fields; Square-shaped Helmholtz coil with electrical current generator; +12 more

Frequency rationale categories: Biological window exploration (26); Black box (20); Power line frequency (20); Low-frequency non-ionizing (18); Thermodynamic model (17); Bioactive window search (17); WPT system operating parameters (16); Based on prior studies (12).

Intensity rationale categories: Regulatory exposure limit (26); Equipment limit (20); Biomolecule electronic perturbation (18); Thermal noise threshold (17); EU exposure limit (17); Exceed ICNIRP limits for testing (16); Based on Crocetti design (12); Workplace exposure limit (10).

Most frequent frequency-rationale details: 26 rows: The study systematically scanned the 20–100 Hz ELF range to identify specific frequencies that reduce viability and proliferation of tumor nerve cell models. Frequencies were selected iteratively by narrowing the scan w... | 18 rows: The equipment uses a non-ionizing multifrequency electromagnetic field with frequency lower than 30 Hz, designed to interact with the electronegative charge of cell

cultures in a cell-type-dependent manner. | 18 rows: 50 Hz was chosen because it is the frequency of ELF magnetic fields generated by electric devices and equipment in clinical practice, industrial environments, and common domestic situations in Europe.

Most frequent intensity-rationale details: 26 rows: 100 μ T corresponds to the exposure limit established by EU Recommendation 1999/519/EC in Spain for a frequency of 50 Hz, chosen as a standard reference intensity for bioelectromagnetic in vitro studies. | 20 rows: Range 0-250 Arms; 250 Arms is the maximum limit of the device, which yields 18 mT | 18 rows: The operational electromagnetic flux emitted by the device is greater than 2 Tesla, designed to generate electronic perturbations that may alter the electronic distribution of biomolecules in a cell-type-dependent manne...

4.1.3 Cell fate outcomes

The table follows the requested order: viability, proliferation, apoptosis, necrosis, ferroptosis, senescence, and motility. Desired directions are interpreted separately for cancer and healthy rows; counterproductive rows are cancer rows with pro-survival, pro-proliferative, or pro-migratory direction.

Table 4.1c. Cell-fate outcomes for ELF/LF magnetic-field platforms.

Outcome	Measured rows	Direction distribution	Strongest desirable cancer effect	Cancer counter-signal	Matched selectivity cases
Viability	109 cancer / 48 healthy (20 papers)	Cancer -1:80; 0:16; +1:12; NA:1 Healthy -1:14; 0:25; +1:8; NA:1	LF-21 / P5 / U373 MG: 0.0% (assay MTT, exposure 0.0833 h, observation 48 h, value 0%); MEMP (ELF-EMF); Pulsed (multifrequency); <30 Hz; AC >2 T	LF-63 / P41 / MDA-MB-231: 137.0% (assay Cell Titer-Blue Cell Viability Assay, exposure 4 h, observation 92 h, value 137%); ELF-EMF; Sine; 50 Hz; AC 1 mT	mixed selectivity: 12; good selectivity: 8; negligible/unclear selectivity: 7
Proliferation	87 cancer / 29 healthy (20 papers)	Cancer -1:56; 0:15; +1:16 Healthy -1:8; 0:18; +1:3	LF-21 / P5 / U373 MG: 0.0% (assay Colony formation assay, exposure 0.0833 h, observation 336 h, value 0%); MEMP (ELF-EMF); Pulsed (multifrequency); <30 Hz; AC >2 T	LF-31 / P14 / MDA-MB-231: 159.1% (assay Trypan Blue (LUNA II automatic cell counter), exposure 24 h, observation 0 h, value 159.1%); ELF-EMF; Square; 45 Hz; AC 100 μ T	good selectivity: 8; mixed selectivity: 7; negligible/unclear selectivity: 3; negative/poor selectivity: 1
Apoptosis	41 cancer / 11 healthy (12 papers)	Cancer -1:1; 0:22; +1:18 Healthy 0:11	LF-55 / P25 / MC4-L2: 914.9% (assay Annexin V-FITC/PI flow cytometry; Western blot (cleaved caspases), exposure 10 h, observation Not reported h, value 914.9%); ELF-EMF; 1 Hz; AC 100 mT	none	negligible/unclear selectivity: 6; good selectivity: 2
Necrosis	18 cancer / 8 healthy (7 papers)	Cancer 0:11; +1:7 Healthy 0:7; +1:1	LF-22 / P6 / U251-P53: 635.0% (assay Flow cytometry (Annexin V-FITC/PI), exposure 10 h, observation 0 h, value 635%); ELF-EMF; 1 Hz; AC 100 mT	none	good selectivity: 2; negligible/unclear selectivity: 2; negative/poor selectivity: 1
Ferroptosis	2 cancer / 0 healthy (1 papers)	Cancer +1:2 Healthy not measured	none	none	none
Senescence	2 cancer / 1 healthy (1 papers)	Cancer +1:2 Healthy 0:1	LF-27 / P10 / MDA-MB-231: 952.0% (assay GL13 (SenTraGor), exposure 5 h, observation 0 h, value 952%); ELF-PEMF (ELF-EMF); Full square (pulsed); 8 Hz; AC 0.011 T	none	good selectivity: 1
Motility	15 cancer / 2 healthy (6 papers)	Cancer -1 only:14; +1 present:1 Healthy -1 only:1; 0 only:1	LF-56 / P28 / 786-O: 53.6% (assay Transwell migration assay, exposure 2.5 h, observation 48 h, value 53.6%); ELF-EMF; Sine; 50 Hz; AC 4.5 mT	none	good selectivity: 1; negative/poor selectivity: 1

Table 4.1d. Matched good-selectivity examples for ELF/LF magnetic-field platforms.

Outcome	Condition ID	Outcome exposure/observation	Cancer rows	Healthy rows
Viability	LF-03	exp/obs 24/0 h	N2A -1 81.41%	C8D1A 1 120.51%
Viability	LF-18	exp/obs 0.5/24 h	U87 -1 60%, LN229 -1 59%, U251 -1 70%	NHA 0 90%, HCF 0 105%, HUVEC 0 96%
Viability	LF-21	exp/obs 0.0417/48 h	U373 MG -1 14%, HeLa -1 24%, MC-38-Luc -1 0%	HaCaT 0 100%, CV-1 0 70%, COS-1 0 115%
Viability	LF-22	exp/obs 10/0 h	U87 -1 81.5%, U251 -1 93.67%, U251-P53 -1 85.87%	Normal human fibroblast 0 99.45%
Viability	LF-27	exp/obs 5/0 h	MDA-MB-231 -1 90%, MCF-7 -1 84%	FF95 1 120%
Viability	LF-48	exp/obs 0.667/48 h	B16-BL6 -1 49.7%, B16-BL6 0 96.57%	HEK293T 0 95%

Outcome	Condition ID	Outcome exposure/observation	Cancer rows	Healthy rows
Viability	LF-56	exp/obs 2.5/72 h	786-O -1 70%, 769-P -1 81%, Caki1 -1 85%	HEK293 1 127%
Viability	LF-59	exp/obs 1/30 h	SH-SY5Y 0 85%, SH-SY5Y -1 66%, T98G 0 113%	HDFa 0 101%, HDFa 0 96%, NHA 0 101%
Proliferation	LF-04	exp/obs 24/0 h	CT2A -1 83.94%, N2A -1 74.01%	C8D1A 1 111.8%
Proliferation	LF-27	exp/obs 5/0 h	MDA-MB-231 -1 16%, MCF-7 -1 33%	FF95 1 123%
Proliferation	LF-31	exp/obs 24/0 h	PC12 -1 75.81%, B16F10 1 136.81%, SKBR3 -1 88.95%	3T3 1 112.53%
Proliferation	LF-33	exp/obs 24/0 h	PC12 1 103.37%, B16F10 1 104.42%, SKBR3 -1 71.74%	3T3 0 90%

Dose/time and risk signals.

Several ELF/LF windows are strong but not monotonic across the whole category. The MEMP condition LF-21 (<30 Hz, >2 T, 5 min) produced 0-24% viability in multiple cancer rows and mostly nonsignificant directions in healthy rows, but the healthy A9/PAM numeric decreases highlight the need to verify sparing beyond Direction alone.

Power-line-like 50 Hz/mT exposures can be either inhibitory or counterproductive: LF-63 (50 Hz, 1 mT, 4 h) increased MDA-MB-231 viability to 137% and produced adverse motility/necrosis selectivity against MCF-10A.

The clearest time-associated oxidative effect is LF-28 (0.01 Hz, 100 mT, 120 h) with DU-145 ROS at 4400%, but HUVEC ROS also reached 1900%, weakening selectivity despite cancer stress.

4.1.4 Mechanistic outcomes

Table 4.1e. Mechanistic coverage and fate linkage for ELF/LF magnetic-field platforms.

Mechanism	Coverage	Observed direction/effect	Same-row links to fate outcomes
ROS	26 cancer / 10 healthy (10 papers); mentioned 18 (2 papers)	Cancer 0:10; +1:16 Healthy 0:3; +1:7	LF-18 U87: Viability -1 (60%), Proliferation -1 (65%); LF-18 LN229: Viability -1 (59%), Proliferation -1 (70%); LF-18 NHA: Viability 0 (90%)
Antioxidant	8 cancer / 5 healthy (4 papers); mentioned 3 (1 papers)	Cancer -1:4; 0:4 Healthy -1:1; 0:4	LF-26 G401: Viability -1 (71%), Proliferation -1 (62.5%), Apoptosis 1 (200%); LF-26 A549: Viability -1 (70%), Proliferation -1 (79%), Apoptosis 1 (275%); LF-60 HDFa: Viability 0 (104%), Apoptosis 0 (134.3%), Necrosis 0 (133.3%)
Membrane Potential	21 cancer / 6 healthy (5 papers); mentioned 10 (4 papers)	Cancer -1:9; 0:7; +1:5 Healthy -1:3; 0:2; +1:1	LF-18 U87: Viability -1 (60%), Proliferation -1 (65%); LF-36 293T: Proliferation 0 (95.5%); LF-36 293T: Proliferation 0 (101%)
Calcium	15 cancer / 3 healthy (4 papers); mentioned 32 (5 papers)	Measured positive/effect: 5; no effect: 13	LF-36 293T: Proliferation 0 (95.5%); LF-36 293T: Proliferation 0 (101%); LF-36 HepG2: Proliferation -1 (82%)
Cellular Morphology	34 cancer / 14 healthy (13 papers); mentioned 0 (0 papers)	Measured positive/effect: 31; no effect: 17	LF-22 U87: Viability -1 (81.5%), Proliferation -1 (83.33%), Apoptosis 1 (255.81%); LF-22 U251: Viability -1 (93.67%), Proliferation -1 (68.42%), Apoptosis 1 (198%); LF-22 U251-P53: Viability -1 (85.87%), Proliferation 0 (69.69%), Apoptosis 1 (598.11%)
Cytoskeleton	16 cancer / 8 healthy (5 papers); mentioned 9 (4 papers)	Measured positive/effect: 15; no effect: 9	LF-53 MDA-MB-231: Viability -1 (not reported%), Proliferation -1 (not reported%), Motility -1 (not reported%); LF-50 MDA-MB-231: Motility -1 (not reported%); LF-51 MDA-MB-231: Motility -1 (not reported%)
Metabolic Reprogramming	10 cancer / 1 healthy (3 papers); mentioned 26 (3 papers)	Measured positive/effect: 10; no effect: 1	LF-09 PANC-1: Proliferation -1 (83%); LF-10 PANC-1: Proliferation -1 (78%); LF-11 1627: Proliferation -1 (80%)
Differentiation	1 cancer / 0 healthy (1 papers); mentioned 2 (1 papers)	Measured positive/effect: 1; no effect: 0	LF-28 K562: Viability -1 (80%), Proliferation -1 (44%), Apoptosis 0 (125%)
Cell Cycle	27 cancer / 16 healthy (8 papers); mentioned 23 (5 papers)	Measured positive/effect: 23; no effect: 32	LF-18 U87: Viability -1 (60%), Proliferation -1 (65%); LF-18 NHA: Viability 0 (90%); LF-21 U373 MG: Viability -1 (14%), Proliferation -1 (20%)

The subgroup suggests at least three partially separable mechanisms: (i) high-amplitude or pulsed ELF/MEMP-like exposures causing loss of proliferation, cell-cycle disruption, and in some cases senescence; (ii) 1 Hz / 100 mT exposures producing apoptosis/necrosis with normal fibroblast sparing; and (iii) oxidative or redox-mitochondrial effects that are strong in selected cancer cells but not universally selective because some healthy cells also show ROS elevation. Motility inhibition appears most coherent when low-mT 50 Hz fields affect renal carcinoma migration without HEK293 migration impairment.

4.1.5 Molecular outcomes

Table 4.1f. Genomic and proteomic findings for ELF/LF magnetic-field platforms.

Block	Coverage	Frequently studied families	Recurrent targets and direction consistency	Cancer/healthy opposite-direction signatures
Genomic	32 conditions, 12 papers; mentioned 0	DNA integrity (8), Mitochondrial / Respiratory (6), Cell cycle regulation (4), Tumor suppressor (2), Apoptosis regulator (2)	DNA strand breaks: n=8, papers=1, dirs={'0': 8}; COX2: n=7, papers=2, dirs={'1': 7}; COX4: n=7, papers=2, dirs={'1': 7}; MT-ATP6: n=7, papers=2, dirs={'1': 7}; ATP5B: n=7, papers=2, dirs={'1': 7}; UCP1: n=4, papers=1, dirs={'1': 4}; UCP2: n=4, papers=1, dirs={'1': 4}; P53: n=4, papers=1, dirs={'1': 3, '-1': 1}	P53: cancer [-1, 1], healthy [1]; P21: cancer [-1, 1], healthy [1]; MDM2: cancer [0, 1], healthy [1]; KLF4: cancer [1], healthy [0]; STAT3: cancer [1], healthy [-1]
Proteomic	18 conditions, 8 papers; mentioned 0	Cell Cycle Control (6), Rho GTPases (4), Glycolysis (3), Mitochondrial Biogenesis (3), Proliferation Marker (3)	Ki67: n=4, papers=2, dirs={'-1': 2, '0': 1, '1': 1}; GTP-Rho: n=4, papers=1, dirs={'-1': 4}; MnSOD: n=3, papers=1, dirs={'1': 2, '0': 1}; Cu/ZnSOD: n=3, papers=1, dirs={'0': 3}; PGC1a: n=3, papers=1, dirs={'1': 2, '0': 1}; p21WAF1/Cip1: n=3, papers=1, dirs={'1': 2, '0': 1}; p-p53: n=2, papers=1, dirs={'1': 2}; p21: n=2, papers=1, dirs={'1': 2}	MnSOD: cancer [1], healthy [0]; PGC1a: cancer [1], healthy [0]; Ki67: cancer [-1], healthy [0, 1]; p21WAF1/Cip1: cancer [1], healthy [0]; ITGA6: cancer [1], healthy [-1]

4.1.6 Mechanism-of-action synthesis within subgroup

Table 4.1g. Parameter combinations with multi-endpoint coherence in ELF/LF magnetic-field platforms.

Condition ID	Desired-outcome count	Exposure	Coherent outcome pattern
LF-21	29	MEMF (ELF-EMF); Pulsed (multifrequency); <30 Hz; AC >2 T	U373 MG Viability -1 14%; U373 MG Proliferation -1 20%; U373 MG Viability -1 0%; U373 MG Proliferation -1 0%; HeLa Viability -1 24%; HeLa Proliferation -1 15%
LF-18	17	AMF (LF-EMF); Sine; 227000 Hz; AC 18 mT	U87 Viability -1 60%; U87 Proliferation -1 65%; U87 ROS 1 150%; LN229 Viability -1 59%; LN229 Proliferation -1 70%; LN229 ROS 1 145%
LF-60	16	EMF (LF-EMF); Sine; 87000 Hz; AC 1.6 mT	HDFA Viability 0 104%; HDFA Apoptosis 0 134.3%; HDFA Necrosis 0 133.3%; HDFA Antioxidant 0 not reported%; HDFA Viability 0 103%; HDFA Apoptosis 0 128.6%
LF-22	16	ELF-EMF; 1 Hz; AC 100 mT	U87 Viability -1 81.5%; U87 Proliferation -1 83.33%; U87 Apoptosis 1 255.81%; U87 Necrosis 1 336%; U251 Viability -1 93.67%; U251 Proliferation -1 68.42%
LF-56	14	ELF-EMF; Sine; 50 Hz; AC 4.5 mT	HEK293 Viability 1 127%; HEK293 Apoptosis 0 not reported%; HEK293 Necrosis 0 not reported%; HEK293 Motility 0 100%; 786-O Viability -1 70%; 786-O Apoptosis 1 not reported%

Interpretation: within this subgroup, the strongest monotherapy signal should be assigned to conditions where cancer-cell fate and mechanistic stress move together while healthy rows remain directionally neutral. Conditions with strong cancer effects but absent healthy controls are hypothesis-generating rather than selective evidence.

4.1.7 Limitations of subgroup evidence base

The apparent strength of this category is partly diluted by heterogeneity: AC intensity ranges from microtesla to tesla, frequencies span sub-Hz to hundreds of kHz, and devices differ fundamentally. Some highly effective conditions are driven by one paper or one device. Direction=0 values can still conceal sizeable nonsignificant numeric shifts, and several healthy comparators are not tissue-matched to the cancer cells.

4.2. RF/HF magnetic field + static-field hybrids

4.2.1 Subgroup overview

This subgroup is biologically intriguing but selectivity-poor in the dataset because every row uses HT-1080 fibrosarcoma and no healthy comparator is present. The exposure rationale is unusually mechanistic: MHz frequencies were chosen around Fe-S cluster hyperfine resonances in the mitochondrial electron-transport chain. The retained evidence base is 21 conditions from 2 papers, with a cancer-to-healthy row ratio of 21:0.

Per-study cancer/healthy ratio: P7:7C/0H; P29:14C/0H.

No paper in this subgroup includes both cancer and healthy/non-cancer lines, so direct selectivity cannot be assessed within the dataset.

Dominant tested biological contexts: Fibrosarcoma (21 rows).

Table 4.2a. Cell-line panel for RF/HF magnetic field + static-field hybrids.

Role	Tumour/tissue type	Species/origin	Cell lines (number of conditions)	Rows
Cancer	Fibrosarcoma	Human	HT-1080 (21)	21

4.2.2 Exposure parameters and rationale

The subgroup contains 21 distinct physical exposure signatures after collapsing rows with identical field parameters. Full condition IDs are provided in Appendix A and are used throughout this module.

Table 4.2b. Parameter envelope for RF/HF magnetic field + static-field hybrids.

Parameter	Values present in retained non-thermal rows
Field labels	RF-MF + SMF (MF/HF-EMF + SMF)
Waveforms	Frequency modulated at 50 kHz
AC frequencies	numeric envelope 900000-7200000; 2000000 Hz; 2500000 Hz; 3000000 Hz; 3500000 Hz; 4000000 Hz; 4500000 Hz; 5000000 Hz; 1800000 Hz; 3600000 Hz; 7200000 Hz; 9000000 Hz
AC magnetic intensity	numeric envelope 0.1-20; 20 nT; 1.5 μ T; 1 nT; 0.1 μ T
DC/static field	numeric envelope 10-300; 45 μ T; 10 μ T; 300 μ T; 90 μ T
Devices	Square Helmholtz coils (30 turns each, side=15 cm, 1.63 mm wire) within a mu-metal shielding box (MuShield Company); RF generated by 15 MHz Function/Arbitrary Waveform Generator (Agilent 33120A); SMF by Triple Output DC Power Supply (Agilent E3631A); measured with fluxgate magnetometer (Walker Scientific FGM-4D2N) and passive loop antenna (Beehive Electronics 100C); ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite plate exposure system with permanent magnets; Mu-metal shielding box

Frequency rationale categories: Hyperfine resonances of Fe-S clusters (14); Hyperfine resonance frequencies (7).

Intensity rationale categories: Investigate weak field effects (14); Below safety standards (7).

Most frequent frequency-rationale details: 14 rows: Frequencies were selected based on hyperfine resonances of iron-sulfur (Fe-S) clusters in the electron transport chain of mitochondria, which show hyperfine interactions with ^{57}Fe of the protein between 2.15 MHz and 3.85... | 7 rows: The 2-5 MHz range was chosen based on known hyperfine resonances of iron-sulfur clusters and other key components of the electron transport chain (ETC) within mitochondria, particularly Complex I, to modulate electron t...

Most frequent intensity-rationale details: 14 rows: Different RF field amplitudes ranging from 1 nT to 1.5 μ T were used to comprehensively investigate the relationship between applied field strength, resonant behavior, and biological phenomena. The static magnetic field ... | 7 rows: The 20 nT RF field intensity was chosen to investigate biological effects at intensities significantly below current safety standards (ICNIRP reference level of 61 V/m for occupational exposure in the 2-5 MHz region). T...

4.2.3 Cell fate outcomes

The table follows the requested order: viability, proliferation, apoptosis, necrosis, ferroptosis, senescence, and motility. Desired directions are interpreted separately for cancer and healthy rows; counterproductive rows are cancer rows with pro-survival, pro-proliferative, or pro-migratory direction.

Table 4.2c. Cell-fate outcomes for RF/HF magnetic field + static-field hybrids.

Outcome	Measured rows	Direction distribution	Strongest desirable cancer effect	Cancer counter-signal	Matched selectivity cases
Viability	7 cancer / 0 healthy (1 papers)	Cancer -1:3; 0:1; +1:3 Healthy not measured	RFM-04 / P7 / HT-1080: 91.0% (assay CellTiter 96 Non-Radioactive Cell Proliferation Assay (MTT), exposure 96 h, observation 0 h, value 91%); RF-MF + SMF (MF/HF-EMF + SMF); 3500000 Hz; AC 20 nT; DC 45 μ T	RFM-05 / P7 / HT-1080: 103.0% (assay CellTiter 96 Non-Radioactive Cell Proliferation Assay (MTT), exposure 96 h, observation 0 h, value 103%); RF-MF + SMF (MF/HF-EMF + SMF); 4000000 Hz; AC 20 nT; DC 45 μ T	none
Proliferation	3 cancer / 0 healthy (2 papers)	Cancer +1:3 Healthy not measured	none	RFM-08 / P29 / HT-1080: 133.0% (assay Countess II Automated Cell Counter (trypan blue exclusion), exposure 96 h, observation 0 h, value 133%); RF-MF + SMF (MF/HF-EMF + SMF); Frequency modulated at 50 kHz; 4000000 Hz; AC 20 nT; DC 45 μ T	none
Apoptosis	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Necrosis	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Ferroptosis	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Senescence	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Motility	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none

No good matched cancer/healthy selectivity cases were available in this subgroup.

Dose/time and risk signals.

Frequency-window behavior dominates: RFM-01/RFM-03 show ROS increases at 2-3 MHz, RFM-04 shows the strongest viability decrease at 3.5 MHz (91%), and RFM-05/RFM-06 around 4-4.5 MHz become pro-survival/pro-proliferative with ROS reduction.

4.2.4 Mechanistic outcomes

Table 4.2e. Mechanistic coverage and fate linkage for RF/HF magnetic field + static-field hybrids.

Mechanism	Coverage	Observed direction/effect	Same-row links to fate outcomes
ROS	16 cancer / 0 healthy (2 papers); mentioned 0 (0 papers)	Cancer -1:9; 0:5; +1:2 Healthy not measured	RFM-01 HT-1080: Viability -1 (96%); RFM-02 HT-1080: Viability 1 (102.3%); RFM-03 HT-1080: Viability 0 (104%)
Antioxidant	7 cancer / 0 healthy (1 papers); mentioned 0 (0 papers)	Cancer +1:1; mixed [-1, 1]; 2; mixed [-1, 0, 1]; 1; mixed [0, 1]; 2; mixed [-1, 0]; 1 Healthy not measured	RFM-01 HT-1080: Viability -1 (96%); RFM-02 HT-1080: Viability 1 (102.3%); RFM-03 HT-1080: Viability 0 (104%)
Membrane Potential	10 cancer / 0 healthy (1 papers); mentioned 7 (1 papers)	Cancer -1:2; 0:4; +1:4 Healthy not measured	no same-row fate linkage
Calcium	8 cancer / 0 healthy (1 papers); mentioned 0 (0 papers)	Measured positive/effect: 2; no effect: 6	no same-row fate linkage
Cellular Morphology	0 cancer / 0 healthy (0 papers); mentioned 1 (1 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Cytoskeleton	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Metabolic Reprogramming	0 cancer / 0 healthy (0 papers); mentioned 21 (2 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Differentiation	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Cell Cycle	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage

The strongest internal pattern is a narrow frequency-window effect: 2.0-3.0/3.5 MHz conditions show mild viability reduction and/or mitochondrial superoxide increases, whereas around 4.0-4.5 MHz the same cell line shows pro-proliferative or pro-survival outcomes with ROS suppression and compensatory antioxidant behavior. This looks more like a resonance probe of mitochondrial redox state than a ready monotherapy window.

4.2.5 Molecular outcomes

Table 4.2f. Genomic and proteomic findings for RF/HF magnetic field + static-field hybrids.

Block	Coverage	Frequently studied families	Recurrent targets and direction consistency	Cancer/healthy opposite-direction signatures
Genomic	not investigated	none	none	none
Proteomic	not investigated	none	none	none

4.2.6 Mechanism-of-action synthesis within subgroup

Table 4.2g. Parameter combinations with multi-endpoint coherence in RF/HF magnetic field + static-field hybrids.

Condition ID	Desired-outcome count	Exposure	Coherent outcome pattern
RFM-01	2	RF-MF + SMF (MF/HF-EMF + SMF); 2000000 Hz; AC 20 nT; DC 45 μ T	HT-1080 Viability -1 96%; HT-1080 ROS 1 127%

Interpretation: within this subgroup, the strongest monotherapy signal should be assigned to conditions where cancer-cell fate and mechanistic stress move together while healthy rows remain directionally neutral. Conditions with strong cancer effects but absent healthy controls are hypothesis-generating rather than selective evidence.

4.2.7 Limitations of subgroup evidence base

All evidence comes from two papers, one cancer cell line, and zero healthy rows. The pro-proliferative 4 MHz window is a major safety signal. Molecular profiling is absent, and the viability effects are modest relative to other categories.

4.3. Intermediate-frequency alternating electric fields

4.3.1 Subgroup overview

This subgroup contains canonical TTFields-like exposures and a small noncanonical AEF row. It is dominated by breast and biliary/glioma models and is one of the clearest selectivity categories because MDA-MB-231 and MCF-12 are repeatedly compared under matched fields. The retained evidence base is 52 conditions from 6 papers, with a cancer-to-healthy row ratio of 41:11.

Per-study cancer/healthy ratio: P17:10C/0H; P26:6C/6H; P27:3C/0H; P34:12C/5H; P39:1C/0H; P49:9C/0H.

Direct selectivity is possible in papers with both cancer and healthy/non-cancer lines: P26, P34.

Dominant tested biological contexts: Triple-negative breast cancer (TNBC) (18 rows); Glioblastoma (12 rows); Breast epithelial cells (11 rows); Intrahepatic cholangiocarcinoma (10 rows); Melanoma (1 rows).

Table 4.3a. Cell-line panel for Intermediate-frequency alternating electric fields.

Role	Tumour/tissue type	Species/origin	Cell lines (number of conditions)	Rows
Cancer	Triple-negative breast cancer (TNBC)	Human	MDA-MB-231 (13), HCC38 (5)	18
Cancer	Glioblastoma	Human	U251 (9), U87 (3)	12
Cancer	Intrahepatic cholangiocarcinoma	Human	HCCC-9810 (5), RBE (5)	10
Cancer	Melanoma	Mouse	B16F10 (1)	1
Healthy/non-cancer	Breast epithelial cells	Human	MCF-12 (11)	11

4.3.2 Exposure parameters and rationale

The subgroup contains 19 distinct physical exposure signatures after collapsing rows with identical field parameters. Full condition IDs are provided in Appendix A and are used throughout this module.

Table 4.3b. Parameter envelope for Intermediate-frequency alternating electric fields.

Parameter	Values present in retained non-thermal rows
Field labels	TTFields (AEF); Tumor Treating Induced Fields (TTIF); AEF
Waveforms	Sine; Sine (UM); Sine (FM)
AC frequencies	numeric envelope 100000-300000; 100000 Hz; 150000 Hz; 200000 Hz; 300000 Hz
Electric field	numeric envelope 0.26-6; 2.1 V/cm; 1.3 V/cm; 1.7 V/cm; 1.5 V/cm; 3 V/cm; 3.75 V/cm; 4.4 V/cm; 6 V/cm; 0.26 V/cm; 2.2 V/cm; 1 V/cm
Devices	TTFields delivery system (Healthy Life Innovation Medical Technology, Jiangsu, China); Custom device: Tektronix AFG1062 function generator connected to a 4-channel relay module, Arduino Uno Rev3 SMD control board, paired stainless steel electrodes fitted into a 6-well cell culture plate lid; TTIF device: motor, wires, capacitor, and magnetic ring forming an LC resonance circuit (transformer-like electric fields device); In-house built TTField delivery device (Tektronix AFG1062 function generator, Arduino Uno Rev3 control board, 4-channel relay module, stainless steel electrodes in 6-well plate lid); Rotary triboelectric nanogenerator (TENG) based on 80 micrometer-thick PTFE film and copper electrodes on 80 mm diameter acrylic substrate, driven by a stepping motor; TEFT system (CL-301A, Antai Kangcheng Biotechnology Co., Ltd.)

Frequency rationale categories: Optimal frequency screening (19); Based on prior studies (17); Prior research findings (12); TTFields GBM sensitivity (3); Low frequency TENG output (1).

Intensity rationale categories: Exploring therapeutic window (17); Prior efficacy studies (12); Dose-response evaluation (10); Dose-dependent efficacy (9); Prior studies reference (3); TENG output characteristics (1).

Most frequent frequency-rationale details: 17 rows: The frequency of 150 kHz was chosen based on combining treatment parameters from Kirson et al. and Berkelmann et al., which are established protocols for TTField delivery. | 12 rows: Human breast carcinoma cells are most affected by field frequencies of 150 kHz with a field intensity of 1.75-3 V/cm, as established in prior studies by Kirson et al. and Smothers et al. | 10 rows: TTFields exhibit potent antitumor activity across a frequency range of 100-300 kHz with optimal frequencies being cell type-dependent. Three representative frequencies (100, 150, 200 kHz) were tested to determine the op...

Most frequent intensity-rationale details: 17 rows: The authors investigated a broad range of electric field intensities (1.5, 3, 3.75, 4.4, and 6 V/cm) to identify the optimal therapeutic window for TNBC treatment and to assess selectivity between cancer and non-cancer ... | 12 rows: Based on previous experimentation, field amplitudes within 1.5-3.0 V/cm showed the greatest treatment selectivity for TNBC cells without destroying epithelial cells. | 10 rows: The relationship between TTFIELDS intensity and BTC cell proliferation was evaluated at intensities of 1.3, 1.7, and 2.1 V/cm to establish a dose-dependent response, consistent with the typical TTFIELDS intensity range ...

4.3.3 Cell fate outcomes

The table follows the requested order: viability, proliferation, apoptosis, necrosis, ferroptosis, senescence, and motility. Desired directions are interpreted separately for cancer and healthy rows; counterproductive rows are cancer rows with pro-survival, pro-proliferative, or pro-migratory direction.

Table 4.3c. Cell-fate outcomes for Intermediate-frequency alternating electric fields.

Outcome	Measured rows	Direction distribution	Strongest desirable cancer effect	Cancer counter-signal	Matched selectivity cases
Viability	24 cancer / 11 healthy (5 papers)	Cancer -1:12; 0:7; +1:2; NA:3 Healthy -1:4; 0:7	AEF-06 / P26 / MDA-MB-231: 0.0% (assay Flow cytometry viable cell counts (ORFLO Moxi Go II), exposure 72 h, observation 0 h, value 0%); TTFIELDS (AEF); Sine (FM); 150000 Hz; E 1.5 V/cm	AEF-04 / P17 / HCCC-9810: 327.4% (assay LDH cytotoxicity assay, exposure 96 h, observation 0 h, value 327.4%); TTFIELDS (AEF); Sine; 150000 Hz; E 2.1 V/cm	good selectivity: 5; mixed selectivity: 3; negligible/unclear selectivity: 2; negative/poor selectivity: 1
Proliferation	19 cancer / 0 healthy (2 papers)	Cancer -1:17; 0:2 Healthy not measured	AEF-04 / P17 / RBE: 48.0% (assay Automated cell counting (Scepter 3.0), exposure 96 h, observation 0 h, value 48%); TTFIELDS (AEF); Sine; 150000 Hz; E 2.1 V/cm	none	none
Apoptosis	3 cancer / 2 healthy (2 papers)	Cancer +1:3 Healthy 0:2	none	none	good selectivity: 2
Necrosis	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Ferroptosis	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Senescence	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Motility	5 cancer / 3 healthy (3 papers)	Cancer -1 only:5 Healthy 0 only:3	AEF-06 / P26 / MDA-MB-231: 19.6% (assay Scratch test (wound healing assay), exposure 24 h, observation 0 h, value 19.6%); TTFIELDS (AEF); Sine (FM); 150000 Hz; E 1.5 V/cm	none	good selectivity: 3

Table 4.3d. Matched good-selectivity examples for Intermediate-frequency alternating electric fields.

Outcome	Condition ID	Outcome exposure/observation	Cancer rows	Healthy rows
Viability	AEF-07	exp/obs 24/0 h	MDA-MB-231 -1 29%	MCF-12 0 89.76%
Viability	AEF-06	exp/obs 48/0 h	MDA-MB-231 -1 51%	MCF-12 0 102%
Viability	AEF-10	exp/obs Not reported/0 h	MDA-MB-231 -1 not reported%, HCC38 -1 not reported%	MCF-12 0 not reported%
Viability	AEF-11	exp/obs Not reported/0 h	MDA-MB-231 0 not reported%, HCC38 -1 not reported%	MCF-12 0 not reported%
Viability	AEF-09	exp/obs Not reported/0 h	HCC38 -1 not reported%	MCF-12 0 not reported%
Apoptosis	AEF-07	exp/obs 24/0 h	MDA-MB-231 1 not reported%	MCF-12 0 not reported%
Apoptosis	AEF-06	exp/obs 24/0 h	MDA-MB-231 1 not reported%	MCF-12 0 not reported%
Motility	AEF-07	exp/obs 24/0 h	MDA-MB-231 -1 30.4%	MCF-12 0 100%
Motility	AEF-06	exp/obs 24/0 h	MDA-MB-231 -1 19.6%	MCF-12 0 93%
Motility	AEF-09	exp/obs 24/0 h	MDA-MB-231 -1 30.4%	MCF-12 0 100%

Dose/time and risk signals.

In MDA-MB-231/MCF-12 matched rows, 150 kHz at 1.5 V/cm shows a better 24-48 h selectivity window than later 72 h viability, where some rows converge toward control. Frequency modulation improves motility inhibition (19.6% of control wound closure) relative to uniform mode (30.4%).

Biliary cancer rows show dose-related proliferation suppression across 1.3-2.1 V/cm, but LDH/cytotoxicity values above control are encoded as viability Direction=1 and should be interpreted as membrane damage/cytotoxicity assay readout rather than improved cancer survival.

4.3.4 Mechanistic outcomes

Table 4.3e. Mechanistic coverage and fate linkage for Intermediate-frequency alternating electric fields.

Mechanism	Coverage	Observed direction/effect	Same-row links to fate outcomes
ROS	1 cancer / 0 healthy (1 papers); mentioned 0 (0 papers)	Cancer +1:1 Healthy not measured	AEF-14 B16F10: Viability -1 (18%), Apoptosis 1 (not reported%)
Antioxidant	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Cancer not measured Healthy not measured	no same-row fate linkage
Membrane Potential	1 cancer / 0 healthy (1 papers); mentioned 0 (0 papers)	Cancer -1:1 Healthy not measured	AEF-14 B16F10: Viability -1 (18%), Apoptosis 1 (not reported%)
Calcium	1 cancer / 0 healthy (1 papers); mentioned 0 (0 papers)	Measured positive/effect: 1; no effect: 0	AEF-14 B16F10: Viability -1 (18%), Apoptosis 1 (not reported%)
Cellular Morphology	13 cancer / 6 healthy (5 papers); mentioned 0 (0 papers)	Measured positive/effect: 12; no effect: 7	AEF-04 HCCC-9810: Viability 1 (327.4%), Proliferation -1 (58.2%), Motility -1 (53.4%); AEF-04 RBE: Viability 1 (229.7%), Proliferation -1 (48%), Motility -1 (27.6%); AEF-07 MDA-MB-231: Viability -1 (29%), Apoptosis 1 (not reported%), Motility -1 (30.4%)
Cytoskeleton	2 cancer / 0 healthy (1 papers); mentioned 9 (1 papers)	Measured positive/effect: 2; no effect: 0	AEF-03 HCCC-9810: Proliferation -1 (72%); AEF-03 RBE: Proliferation -1 (52%)
Metabolic Reprogramming	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Differentiation	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Cell Cycle	0 cancer / 0 healthy (0 papers); mentioned 20 (3 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage

The strongest coherent profile is mitosis/motility vulnerability: 150 kHz, 1.5 V/cm TTFields, especially frequency modulation, suppresses MDA-MB-231 viability and wound closure while sparing MCF-12 motility and often viability. The mechanistic data are thinner than the phenotype, but morphology/cytoskeleton and DAMP/ICD proteins in the AEF-related rows suggest a bridge between physical mitotic stress and immunogenic stress signaling.

4.3.5 Molecular outcomes

Table 4.3f. Genomic and proteomic findings for Intermediate-frequency alternating electric fields.

Block	Coverage	Frequently studied families	Recurrent targets and direction consistency	Cancer/healthy opposite-direction signatures
Genomic	not investigated	none	none	none
Proteomic	2 conditions, 1 papers; mentioned 0	Damage-associated molecular patterns (DAMPs) / Immunogenic cell death (ICD) biomarkers (2)	CRT: n=2, papers=1, dirs={'1': 2}; HMGB1: n=2, papers=1, dirs={'1': 2}; ATP: n=2, papers=1, dirs={'1': 2}	none detected

4.3.6 Mechanism-of-action synthesis within subgroup

Table 4.3g. Parameter combinations with multi-endpoint coherence in Intermediate-frequency alternating electric fields.

Condition ID	Desired-outcome count	Exposure	Coherent outcome pattern
AEF-07	8	TTFields (AEF); Sine (UM); 150000 Hz; E 1.5 V/cm	MDA-MB-231 Viability -1 29%; MDA-MB-231 Apoptosis 1 not reported%; MDA-MB-231 Motility -1 30.4%; MDA-MB-231 Viability -1 74%; MCF-12 Viability 0 89.76%; MCF-12 Apoptosis 0 not reported%
AEF-06	8	TTFields (AEF); Sine (FM); 150000 Hz; E 1.5 V/cm	MDA-MB-231 Viability -1 29%; MDA-MB-231 Apoptosis 1 not reported%; MDA-MB-231 Motility -1 19.6%; MDA-MB-231 Viability -1 51%; MDA-MB-231 Viability -1 0%; MCF-12 Apoptosis 0 not reported%
AEF-09	6	TTFields (AEF); Sine; 150000 Hz; E 1.5 V/cm	MDA-MB-231 Viability -1 30%; MDA-MB-231 Motility -1 30.4%; MDA-MB-231 Viability -1 74%; HCC38 Viability -1 not reported%; MCF-12 Viability 0 not reported%; MCF-12 Motility 0 100%
AEF-18	4	TTFields (AEF); Sine; 200000 Hz; E 2.2 V/cm	U251 Proliferation -1 not reported%; U251 Proliferation -1 not reported%; U251 Proliferation -1 not reported%; U251 Proliferation -1 not reported%
AEF-04	4	TTFields (AEF); Sine; 150000 Hz; E 2.1 V/cm	HCCC-9810 Proliferation -1 58.2%; HCCC-9810 Motility -1 53.4%; RBE Proliferation -1 48%; RBE Motility -1 27.6%

Interpretation: within this subgroup, the strongest monotherapy signal should be assigned to conditions where cancer-cell fate and mechanistic stress move together while healthy rows remain directionally neutral. Conditions with strong cancer effects but absent healthy controls are hypothesis-generating rather than selective evidence.

4.3.7 Limitations of subgroup evidence base

Most conditions measure viability/proliferation/motility but not ROS, calcium, mitochondrial potential, antioxidants, ferroptosis, necrosis, or detailed cell cycle by direct assay. Several rows lack % values despite significant directions. The noncanonical TENG-AEF result is promising but not directly interchangeable with TTFields dosimetry.

4.4. Nanosecond pulsed electric fields

4.4.1 Subgroup overview

The nsPEF subgroup is defined by pulse dose rather than AC frequency. It contains strong cancer-cell killing in 3D glioma spheroids and selective breast-cell effects under MCF-7/MCF-10A comparison, but healthy comparator coverage is limited overall. The retained evidence base is 37 conditions from 5 papers, with a cancer-to-healthy row ratio of 34:3.

Per-study cancer/healthy ratio: P42:21C/0H; P43:1C/0H; P45:3C/3H; P46:3C/0H; P47:6C/0H.

Direct selectivity is possible in papers with both cancer and healthy/non-cancer lines: P45.

Dominant tested biological contexts: Triple-negative breast cancer (TNBC) (21 rows); Breast cancer (3 rows); Breast epithelial cells (3 rows); Glioma (2 rows); Melanotic melanoma (2 rows).

Table 4.4a. Cell-line panel for Nanosecond pulsed electric fields.

Role	Tumour/tissue type	Species/origin	Cell lines (number of conditions)	Rows
Cancer	Triple-negative breast cancer (TNBC)	Human	MDA-MB-231 (21)	21
Cancer	Breast cancer	Human	MCF-7 (3)	3
Cancer	Melanotic melanoma	Human	A375 (2)	2
Cancer	Amelanotic melanoma	Human	C32 (1)	1
Cancer	Glioblastoma	Human	U-87 MG (1)	1
Cancer	Glioma	Rat	C6 (1)	1
Cancer	Glioma	Mouse	GL261 (1)	1
Cancer	Hepatocellular carcinoma	Human	Hep3B (1)	1
Cancer	Malignant melanoma (metastatic)	Human	MeWo (1)	1
Cancer	Melanoma (metastatic)	Human	Me45 (1)	1
Cancer	Metastatic melanoma	Human	Colo-829 (1)	1
Healthy/non-cancer	Breast epithelial cells	Human	MCF-10A (3)	3

4.4.2 Exposure parameters and rationale

The subgroup contains 27 distinct physical exposure signatures after collapsing rows with identical field parameters. Full condition IDs are provided in Appendix A and are used throughout this module.

Table 4.4b. Parameter envelope for Nanosecond pulsed electric fields.

Parameter	Values present in retained non-thermal rows
Field labels	nsPEF
Waveforms	Square
Electric field	numeric envelope 8-100; 80 kV/cm; 70 kV/cm; 90 kV/cm; 100 kV/cm; 10 kV/cm; 20 kV/cm; 15 kV/cm; 40 kV/cm; 8 kV/cm
Pulse duration	numeric envelope 5-300; 5 ns; 300 ns; 50 ns; 100 ns
Pulse repetition rate	numeric envelope 2-200000; 100 Hz; 2 Hz; 2000 Hz; 200000 Hz; 10000 Hz
Number of pulses	numeric envelope 60-10000; 1000; 2000; 4000; 6000; 8000; 10000; 800; 60; 100
Devices	Custom-built nanosecond pulse generator (laboratory-developed), 3-5 kV output, with custom microelectrodes (0.5 mm spacing); Pulse generator device with two electrode needles, provided by Hangzhou RuiDi Biological Technology Co., LTD (China). Waveforms detected with a digital fluorescence oscilloscope (DPO4054, Tektronix, USA). Cells treated in a 0.1 mL colorimeter cuvette.; 3D gold microelectrode device integrated with confocal microscope, pulse generator AVL-5-B-TR (AVTECH); Custom-made pulse generator based on dual Marx circuit topology with 1.7 kV MOSFET switches and custom PCB electrodes with ENIG process; Square wave electroporator (100 ns-1 ms) developed at the Institute of High Magnetic Fields, VGTU, Vilnius, Lithuania

Frequency rationale categories: not reported.

Intensity rationale categories: Intracellular targeting (21); Optimal cancer treatment (6); Sub-electroporation threshold (6); Tumor ablation efficacy (3); Clinical applicability (1).

Most frequent intensity-rationale details: 21 rows: High electric field intensities (70-100 kV/cm) were selected to ensure that the 5 ns pulses, which have rich high-frequency components, could penetrate the cell membrane barrier and directly act on intracellular membran... | 6 rows: 8 kV/cm was chosen as a sub-electroporation threshold intensity that does not induce cytotoxicity or full membrane permeabilization but is sufficient to trigger electro-exocytosis and release

of microvesicles containing... | 3 rows: The amplitude of 40 kV/cm was selected for ultrashort pulsed electric fields applied to the multicellular tumor spheroids to achieve effective tumor cell killing while disrupting the dense structure of glioma tumors.

4.4.3 Cell fate outcomes

The table follows the requested order: viability, proliferation, apoptosis, necrosis, ferroptosis, senescence, and motility. Desired directions are interpreted separately for cancer and healthy rows; counterproductive rows are cancer rows with pro-survival, pro-proliferative, or pro-migratory direction.

Table 4.4c. Cell-fate outcomes for Nanosecond pulsed electric fields.

Outcome	Measured rows	Direction distribution	Strongest desirable cancer effect	Cancer counter-signal	Matched selectivity cases
Viability	12 cancer / 3 healthy (4 papers)	Cancer -1:6; 0:6 Healthy 0:3	NPEF-26 / P46 / C6: 7.0% (assay CellTiter-Lumi luminescent 3D cell viability assay (ATP content measurement), exposure N/A h, observation 3 h, value 7%); nsPEF; E 40 kV/cm; pulse 100 ns; rep 200000 Hz; 60 pulses	none	good selectivity: 2; negligible/unclear selectivity: 1
Proliferation	5 cancer / 0 healthy (3 papers)	Cancer -1:4; 0:1 Healthy not measured	NPEF-26 / P46 / GL261: 45.56% (assay Immunofluorescence (Ki67) and Western Blot (Ki67), exposure N/A h, observation 3 h, value 45.56%); nsPEF; E 40 kV/cm; pulse 100 ns; rep 200000 Hz; 60 pulses	none	none
Apoptosis	4 cancer / 1 healthy (2 papers)	Cancer +1:4 Healthy 0:1	NPEF-26 / P46 / GL261: 340.0% (assay Flow cytometry with Annexin V-EGFP/PI dual staining, exposure N/A h, observation 3 h, value 340%); nsPEF; E 40 kV/cm; pulse 100 ns; rep 200000 Hz; 60 pulses	none	good selectivity: 1
Necrosis	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Ferroptosis	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Senescence	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Motility	2 cancer / 0 healthy (1 papers)	Cancer -1 only:1; 0 only:1 Healthy not measured	none	none	none

Table 4.4d. Matched good-selectivity examples for Nanosecond pulsed electric fields.

Outcome	Condition ID	Outcome exposure/observation	Cancer rows	Healthy rows
Viability	NPEF-25	exp/obs 0.033/0 h	MCF-7 -1 not reported%	MCF-10A 0 not reported%
Viability	NPEF-24	exp/obs 0.033/0 h	MCF-7 -1 not reported%	MCF-10A 0 not reported%
Apoptosis	NPEF-25	exp/obs 0.033/0 h	MCF-7 1 not reported%	MCF-10A 0 not reported%
ROS	NPEF-25	exp/obs 0.083/0 h	MCF-7 1 150%	MCF-10A -1 50%

Dose/time and risk signals.

Pulse dose is the principal axis: in 3D glioma spheroids, 40 kV/cm, 100 ns, 60 pulses reduced ATP viability to 7-9% in C6/U-87 MG and increased apoptosis to 320-340%. In breast cells, 20 kV/cm, 50 ns, 2 kHz was the best matched window because MCF-7 showed ROS/apoptosis and MCF-10A did not.

4.4.4 Mechanistic outcomes

Table 4.4e. Mechanistic coverage and fate linkage for Nanosecond pulsed electric fields.

Mechanism	Coverage	Observed direction/effect	Same-row links to fate outcomes
ROS	1 cancer / 1 healthy (1 papers); mentioned 0 (0 papers)	Cancer +1:1 Healthy -1:1	NPEF-25 MCF-7: Viability -1 (not reported%), Apoptosis 1 (not reported%); NPEF-25 MCF-10A: Viability 0 (not reported%), Apoptosis 0 (not reported%)
Antioxidant	0 cancer / 0 healthy (0 papers); mentioned 5 (1 papers)	Cancer not measured Healthy not measured	no same-row fate linkage
Membrane Potential	21 cancer / 1 healthy (2 papers); mentioned 1 (1 papers)	Cancer -1:21 Healthy -1:1	NPEF-25 MCF-7: Viability -1 (not reported%), Apoptosis 1 (not reported%); NPEF-25 MCF-10A: Viability 0 (not reported%), Apoptosis 0 (not reported%)
Calcium	1 cancer / 1 healthy (1 papers); mentioned 0 (0 papers)	Measured positive/effect: 1; no effect: 1	NPEF-25 MCF-7: Viability -1 (not reported%), Apoptosis 1 (not reported%); NPEF-25 MCF-10A: Viability 0 (not reported%), Apoptosis 0 (not reported%)

Mechanism	Coverage	Observed direction/effect	Same-row links to fate outcomes
Cellular Morphology	28 cancer / 1 healthy (5 papers); mentioned 0 (0 papers)	Measured positive/effect: 28; no effect: 1	NPEF-22 Hep3B: Viability -1 (24.94%), Proliferation -1 (48.74%); NPEF-25 MCF-7: Viability -1 (not reported%), Apoptosis 1 (not reported%); NPEF-25 MCF-10A: Viability 0 (not reported%), Apoptosis 0 (not reported%)
Cytoskeleton	2 cancer / 0 healthy (2 papers); mentioned 23 (2 papers)	Measured positive/effect: 2; no effect: 0	NPEF-27 A375: Viability 0 (not reported%), Proliferation 0 (not reported%), Motility 0 (not reported%)
Metabolic Reprogramming	6 cancer / 1 healthy (2 papers); mentioned 0 (0 papers)	Measured positive/effect: 6; no effect: 1	NPEF-25 MCF-7: Viability -1 (not reported%), Apoptosis 1 (not reported%); NPEF-25 MCF-10A: Viability 0 (not reported%), Apoptosis 0 (not reported%); NPEF-27 A375: Viability 0 (not reported%), Proliferation 0 (not reported%), Motility 0 (not reported%)
Differentiation	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Cell Cycle	0 cancer / 0 healthy (0 papers); mentioned 3 (1 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage

The coherent mechanism is acute membrane/organelle electroporation leading to ATP loss, mitochondrial membrane-potential collapse, ROS elevation, caspase activation, and later viability/proliferation loss. The MCF-7/MCF-10A matched rows are especially important because MCF-7 shows viability reduction/apoptosis and ROS increase while MCF-10A remains directionally preserved or has ROS decrease.

4.4.5 Molecular outcomes

Table 4.4f. Genomic and proteomic findings for Nanosecond pulsed electric fields.

Block	Coverage	Frequently studied families	Recurrent targets and direction consistency	Cancer/healthy opposite-direction signatures
Genomic	3 conditions, 1 papers; mentioned 0	MAGE (Melanoma Antigen Gene) family (3)	MAGEA1: n=3, papers=1, dirs={'1': 2, '0': 1}; MAGEA2: n=3, papers=1, dirs={'1': 2, '0': 1}; MAGEA3/A6: n=3, papers=1, dirs={'0': 2, '1': 1}; MAGEA8: n=3, papers=1, dirs={'0': 3}; MAGEA12: n=3, papers=1, dirs={'0': 3}; MAGEC2: n=3, papers=1, dirs={'0': 3}	none detected
Proteomic	11 conditions, 3 papers; mentioned 0	MAGE (Melanoma Antigen Gene) family (6), PD-1 (immune checkpoint) (5), Adhesion and junction proteins (3), Caspases (executioner) (2)	MAGE-A1: n=6, papers=1, dirs={'1': 3, '0': 3}; MAGE-A2: n=5, papers=1, dirs={'1': 1, '0': 4}; MAGE-A3: n=5, papers=1, dirs={'0': 4, '1': 1}; MAGE-A6: n=5, papers=1, dirs={'0': 4, '1': 1}; MAGE-A8: n=5, papers=1, dirs={'0': 5}; MAGE-A12: n=5, papers=1, dirs={'0': 5}; MAGE-C2: n=5, papers=1, dirs={'0': 5}; PD-1 glycosylated: n=5, papers=1, dirs={'1': 4, '0': 1}	Caspase-3/7: cancer [1], healthy [0]

4.4.6 Mechanism-of-action synthesis within subgroup

Table 4.4g. Parameter combinations with multi-endpoint coherence in Nanosecond pulsed electric fields.

Condition ID	Desired-outcome count	Exposure	Coherent outcome pattern
NPEF-26	9	nsPEF; E 40 kV/cm; pulse 100 ns; rep 200000 Hz; 60 pulses	U-87 MG Viability -1 9%; U-87 MG Proliferation -1 53.17%; U-87 MG Apoptosis 1 320%; C6 Viability -1 7%; C6 Proliferation -1 46.81%; C6 Apoptosis 1 330%
NPEF-25	6	nsPEF; E 20 kV/cm; pulse 50 ns; rep 2000 Hz	MCF-7 Viability -1 not reported%; MCF-7 Apoptosis 1 not reported%; MCF-7 ROS 1 150%; MCF-10A Viability 0 not reported%; MCF-10A Apoptosis 0 not reported%; MCF-10A ROS -1 50%
NPEF-24	2	nsPEF; E 15 kV/cm; pulse 50 ns; rep 2000 Hz	MCF-7 Viability -1 not reported%; MCF-10A Viability 0 not reported%
NPEF-22	2	nsPEF; E 10 kV/cm; pulse 300 ns; rep 2 Hz; 800 pulses	Hep3B Viability -1 24.94%; Hep3B Proliferation -1 48.74%

Interpretation: within this subgroup, the strongest monotherapy signal should be assigned to conditions where cancer-cell fate and mechanistic stress move together while healthy rows remain directionally neutral. Conditions with strong cancer effects but absent healthy controls are hypothesis-generating rather than selective evidence.

4.4.7 Limitations of subgroup evidence base

Many strongest effects are in cancer-only rows. The 3D spheroid ATP assay may overstate cell death because ATP leakage after electroporation can precede irreversible death. Pulse parameters vary widely, so intensity, pulse number, repetition rate, and pulse duration are confounded.

4.5. Ultrashort pulsed magnetic fields

4.5.1 Subgroup overview

This subgroup is methodologically narrow: all rows are from one C6 rat glioma study. It cleanly maps a magnetic pulse-parameter window, but it cannot support selectivity because no healthy line was tested. The retained evidence base is 24 conditions from 1 papers, with a cancer-to-healthy row ratio of 24:0.

Per-study cancer/healthy ratio: P22:24C/0H.

No paper in this subgroup includes both cancer and healthy/non-cancer lines, so direct selectivity cannot be assessed within the dataset.

Dominant tested biological contexts: Glioma (24 rows).

Table 4.5a. Cell-line panel for Ultrashort pulsed magnetic fields.

Role	Tumour/tissue type	Species/origin	Cell lines (number of conditions)	Rows
Cancer	Glioma	Rat	C6 (24)	24

4.5.2 Exposure parameters and rationale

The subgroup contains 24 distinct physical exposure signatures after collapsing rows with identical field parameters. Full condition IDs are provided in Appendix A and are used throughout this module.

Table 4.5b. Parameter envelope for Ultrashort pulsed magnetic fields.

Parameter	Values present in retained non-thermal rows
Field labels	nsPMF; μ sPMF
Waveforms	Square; Exponential
AC magnetic intensity	numeric envelope 0.24-854; 0.24 mT; 0.353 mT; 0.598 mT; 0.787 mT; 311 mT; 151 mT; 35.9 mT; 16.4 mT; 122 mT; 363 mT; 854 mT; 45.4 mT
Pulse duration	numeric envelope 100-600; 600 ns; 205 μ s; 100 μ s
Pulse repetition rate	numeric envelope 1-20; 1 Hz; 5 Hz; 10 Hz; 20 Hz
Devices	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial cable; Solenoid-based pulsed electromagnetic field generation device with square wave or exponential wave current source

Frequency rationale categories: not reported.

Intensity rationale categories: Window effect investigation (24).

Most frequent intensity-rationale details: 24 rows: Multiple magnetic field intensities were tested to identify the effective window of action of the magnetic field parameters for inhibiting cancer cell viability, including rise rate dB/dt, intensity B, action integral S...

4.5.3 Cell fate outcomes

The table follows the requested order: viability, proliferation, apoptosis, necrosis, ferroptosis, senescence, and motility. Desired directions are interpreted separately for cancer and healthy rows; counterproductive rows are cancer rows with pro-survival, pro-proliferative, or pro-migratory direction.

Table 4.5c. Cell-fate outcomes for Ultrashort pulsed magnetic fields.

Outcome	Measured rows	Direction distribution	Strongest desirable cancer effect	Cancer counter-signal	Matched selectivity cases
Viability	24 cancer / 0 healthy (1 papers)	Cancer -1:16; 0:8 Healthy not measured	PMF-21 / P22 / C6: 67.47% (assay MTT assay, exposure 16 h, observation 0 h, value 67.47%); μ sPMF; Square; AC 122 mT; pulse 100 μ s	none	none
Proliferation	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Apoptosis	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Necrosis	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none

Outcome	Measured rows	Direction distribution	Strongest desirable cancer effect	Cancer counter-signal	Matched selectivity cases
Ferroptosis	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Senescence	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Motility	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none

No good matched cancer/healthy selectivity cases were available in this subgroup.

Dose/time and risk signals.

The single C6 paper shows a nonmonotonic window: 122 mT square 100 us pulses were more inhibitory than the higher 363 mT setting, indicating pulse waveform/dB/dt and cumulative exposure time rather than peak intensity alone.

4.5.4 Mechanistic outcomes

Table 4.5e. Mechanistic coverage and fate linkage for Ultrashort pulsed magnetic fields.

Mechanism	Coverage	Observed direction/effect	Same-row links to fate outcomes
ROS	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Cancer not measured Healthy not measured	no same-row fate linkage
Antioxidant	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Cancer not measured Healthy not measured	no same-row fate linkage
Membrane Potential	0 cancer / 0 healthy (0 papers); mentioned 24 (1 papers)	Cancer not measured Healthy not measured	no same-row fate linkage
Calcium	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Cellular Morphology	24 cancer / 0 healthy (1 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 24	PMF-01 C6: Viability -1 (84.97%); PMF-05 C6: Viability -1 (85.9%); PMF-09 C6: Viability -1 (85.29%)
Cytoskeleton	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Metabolic Reprogramming	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Differentiation	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Cell Cycle	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage

Viability inhibition depends on waveform, pulse duration, intensity, and cumulative exposure time, with square 100 us pulses at 122 mT producing the strongest C6 reduction. Morphology was measured in all rows; proliferation and membrane-potential involvement are mentioned rather than directly measured.

4.5.5 Molecular outcomes

Table 4.5f. Genomic and proteomic findings for Ultrashort pulsed magnetic fields.

Block	Coverage	Frequently studied families	Recurrent targets and direction consistency	Cancer/healthy opposite-direction signatures
Genomic	not investigated	none	none	none
Proteomic	not investigated	none	none	none

4.5.6 Mechanism-of-action synthesis within subgroup

No condition accumulated multiple measured desired outcomes in this subgroup.

Interpretation: within this subgroup, the strongest monotherapy signal should be assigned to conditions where cancer-cell fate and mechanistic stress move together while healthy rows remain directionally neutral. Conditions with strong cancer effects but absent healthy controls are hypothesis-generating rather than selective evidence.

4.5.7 Limitations of subgroup evidence base

One paper, one cancer line, no healthy comparator, no genomic/proteomic data, and no direct apoptosis/ROS/cell-cycle measurements. The category is useful for parameter-window hypothesis generation, not for therapeutic selectivity conclusions.

4.6. RF/mobile-frequency electromagnetic exposures

4.6.1 Subgroup overview

This subgroup mixes LTE/SAR exposures, radioelectric conveyance, and 13.56 MHz RF. Cancer-cell proliferation suppression under LTE can be large, but the same SAR levels also reduce proliferation in healthy ASC/IMR-90 in some conditions, weakening selectivity. The retained evidence base is 38 conditions from 4 papers, with a cancer-to-healthy row ratio of 31:7.

Per-study cancer/healthy ratio: P30:3C/0H; P31:8C/4H; P35:14C/3H; P37:6C/0H.

Direct selectivity is possible in papers with both cancer and healthy/non-cancer lines: P31, P35.

Dominant tested biological contexts: Hepatocellular carcinoma (8 rows); Cervical adenocarcinoma (6 rows); Neuroblastoma (6 rows); Adipose tissue-derived stem cells (5 rows); Breast cancer (3 rows).

Table 4.6a. Cell-line panel for RF/mobile-frequency electromagnetic exposures.

Role	Tumour/tissue type	Species/origin	Cell lines (number of conditions)	Rows
Cancer	Hepatocellular carcinoma	Human	Huh7 (8)	8
Cancer	Cervical adenocarcinoma	Human	HeLa (6)	6
Cancer	Neuroblastoma	Rat	B103 (4)	4
Cancer	Breast cancer	Human	MCF-7 (3)	3
Cancer	Colorectal adenocarcinoma (left supraclavicular metastasis)	Human	LoVo (2)	2
Cancer	Colorectal adenocarcinoma (primary, Duke's B)	Human	SW480 (2)	2
Cancer	Colorectal carcinoma (primary tumor)	Human	HT29 (2)	2
Cancer	Liver cancer stem cells (hepatocellular carcinoma)	Human	Hep3B (2)	2
Cancer	Neuroblastoma	Human	SH-SY5Y (2)	2
Healthy/non-cancer	Adipose tissue-derived stem cells	Human	ASC (5)	5
Healthy/non-cancer	Normal fibroblast	Human	IMR-90 (2)	2

4.6.2 Exposure parameters and rationale

The subgroup contains 9 distinct physical exposure signatures after collapsing rows with identical field parameters. Full condition IDs are provided in Appendix A and are used throughout this module.

Table 4.6b. Parameter envelope for RF/mobile-frequency electromagnetic exposures.

Parameter	Values present in retained non-thermal rows
Field labels	REAC (RF-EMF); LTE (UHF-EMF); RF-EMF (MF-EMF)
Waveforms	LTE (WCDMA modulation); Continuous sine; Amplitude-modulated sine
AC frequencies	numeric envelope 13560000-17600000000; 17000000000 Hz; 17600000000 Hz; 135600000 Hz
SAR	numeric envelope 0.4-8; 1; 2; 0.4; 4; 8
Devices	BENE 110 (ASMED, Italy); Radioelectric asymmetric conveyance; Radial transmission line (RTL) exposure system with conical antenna, installed in incubator with forced refrigerated water-cooling system; Radial transmission line (RTL) exposure system with E4438C signal generator (Keysight Technologies), customized power amplifier (60 W max), E4418B power meter, conical antenna, JS-CO2-AT-750 incubator, and C-WBL water circulator; LabEHY-200 (Oncotherm Kft., Budapest, Hungary) with novel in vitro applicator

Frequency rationale categories: LTE 4G telecommunications (17); Mobile communication frequency (12); Clinical hyperthermia frequency (6); Restore bioelectric activity (3).

Intensity rationale categories: ICNIRP safety limits testing (17); Regulatory exposure limits (12); Standard preclinical power (6); Endogenous charge asymmetry restoration (3).

Most frequent frequency-rationale details: 17 rows: 1.7 GHz LTE technology in the 1.7 to 1.8 GHz range is widely used in 4th generation mobile technologies, especially in Korea, making its biological effects a major public health concern. | 12 rows: 1700 to 1950 MHz RF-EMFs are the most widely used frequencies in mobile communications, and 4G-LTE technology effects on human cells have not been well documented. | 6 rows: 13.56 MHz is an ISM band frequency used

in clinical capacitive hyperthermia systems (e.g., EHY-2000+) and was selected to enable direct translational comparison with clinical modulated electro-hyperthermia devices.

Most frequent intensity-rationale details: 17 rows: The ICNIRP defines a SAR of 2 W/kg for local head/torso as the safety limit. The study tested SAR values ranging from 0.4 to 8 W/Kg to evaluate effects at and above the regulatory safety limit. | 12 rows: Tested at 1 W/kg and 2 W/kg SAR considering the current maximum permitted exposure values (2 W/kg in Europe and 1.6 W/kg in the US). | 6 rows: An adjusted power of $P_{adj} = 10$ W was used for the in vitro applicator. Specific electric or magnetic field intensities were not explicitly reported for the in vitro setup. An E-field dose of approximately 60,000 V·min/m...

4.6.3 Cell fate outcomes

The table follows the requested order: viability, proliferation, apoptosis, necrosis, ferroptosis, senescence, and motility. Desired directions are interpreted separately for cancer and healthy rows; counterproductive rows are cancer rows with pro-survival, pro-proliferative, or pro-migratory direction.

Table 4.6c. Cell-fate outcomes for RF/mobile-frequency electromagnetic exposures.

Outcome	Measured rows	Direction distribution	Strongest desirable cancer effect	Cancer counter-signal	Matched selectivity cases
Viability	15 cancer / 1 healthy (2 papers)	Cancer -1:3; 0:12 Healthy 0:1	RF-01 / P30 / MCF-7: 84.53% (assay Trypan Blue, exposure 168 h, observation 168 h, value 84.53%); REAC (RF-EMF)	none	negligible/unclear selectivity: 1
Proliferation	10 cancer / 6 healthy (2 papers)	Cancer -1:8; 0:2 Healthy -1:3; 0:3	RF-03 / P31 / SH-SY5Y: 12.0% (assay Cell counting (Cellometer Auto T4, Nexcelom Bioscience), exposure 72 h, observation 0 h, value 12%); LTE (UHF-EMF); LTE (WCDMA modulation); 1700000000 Hz; SAR 2 W/kg	none	mixed selectivity: 2; negligible/unclear selectivity: 2
Apoptosis	10 cancer / 3 healthy (3 papers)	Cancer 0:5; +1:5 Healthy 0:3	RF-09 / P37 / SW480: 145.0% (assay FACS (Annexin V-positive cells), exposure 1 h, observation 0 h, value 145%); RF-EMF (MF-EMF); Continuous sine; 13560000 Hz	none	negligible/unclear selectivity: 3
Necrosis	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Ferroptosis	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Senescence	5 cancer / 2 healthy (2 papers)	Cancer 0:2; +1:3 Healthy +1:2	RF-01 / P30 / MCF-7: 133.0% (assay SA-beta-Galactosidase, exposure 168 h, observation 168 h, value 133%); REAC (RF-EMF)	none	negative/poor selectivity: 1; mixed selectivity: 1
Motility	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none

No good matched cancer/healthy selectivity cases were available in this subgroup.

Dose/time and risk signals.

LTE at 2 W/kg produced strong cancer proliferation suppression, but ASC and IMR-90 were also suppressed at 1-2 W/kg. This prevents a clean monotherapy window at the tested SARs despite large cancer-cell effects.

4.6.4 Mechanistic outcomes

Table 4.6e. Mechanistic coverage and fate linkage for RF/mobile-frequency electromagnetic exposures.

Mechanism	Coverage	Observed direction/effect	Same-row links to fate outcomes
ROS	2 cancer / 2 healthy (1 papers); mentioned 0 (0 papers)	Cancer +1:2 Healthy +1:2	RF-02 ASC: Proliferation 0 (88%), Apoptosis 0 (not reported%), Senescence 1 (not reported%); RF-03 ASC: Proliferation -1 (46%), Apoptosis 0 (not reported%), Senescence 1 (not reported%); RF-02 Huh7: Proliferation -1 (79%), Apoptosis 0 (not reported%), Senescence 0 (not reported%)
Antioxidant	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Cancer not measured Healthy not measured	no same-row fate linkage
Membrane Potential	0 cancer / 0 healthy (0 papers); mentioned 6 (1 papers)	Cancer not measured Healthy not measured	no same-row fate linkage

Mechanism	Coverage	Observed direction/effect	Same-row links to fate outcomes
Calcium	0 cancer / 0 healthy (0 papers); mentioned 6 (1 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Cellular Morphology	3 cancer / 0 healthy (1 papers); mentioned 0 (0 papers)	Measured positive/effect: 3; no effect: 0	RF-01 MCF-7: Viability -1 (95%), Senescence 1 (116.6%); RF-01 MCF-7: Viability -1 (96.7%), Senescence 0 (111%); RF-01 MCF-7: Viability -1 (84.53%), Senescence 1 (133%)
Cytoskeleton	0 cancer / 0 healthy (0 papers); mentioned 6 (1 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Metabolic Reprogramming	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Differentiation	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Cell Cycle	4 cancer / 3 healthy (2 papers); mentioned 3 (1 papers)	Measured positive/effect: 3; no effect: 5	RF-02 ASC: Proliferation 0 (88%), Apoptosis 0 (not reported%), Senescence 1 (not reported%); RF-03 ASC: Proliferation -1 (46%), Apoptosis 0 (not reported%), Senescence 1 (not reported%); RF-02 Huh7: Proliferation -1 (79%), Apoptosis 0 (not reported%), Senescence 0 (not reported%)

The data suggest RF can affect proliferation, senescence, apoptosis, and DNA-stress markers, but the mechanisms are inconsistent across devices. LTE produces ROS in both cancer and healthy cells under matched conditions; 13.56 MHz RF induces apoptosis in colorectal lines without healthy comparators; REAC causes mild MCF-7 viability loss and senescence over long exposure.

4.6.5 Molecular outcomes

Table 4.6f. Genomic and proteomic findings for RF/mobile-frequency electromagnetic exposures.

Block	Coverage	Frequently studied families	Recurrent targets and direction consistency	Cancer/healthy opposite-direction signatures
Genomic	3 conditions, 1 papers; mentioned 0	Stemness (3), Cell cycle / Senescence (3), Apoptosis / Stress response (3)	Oct-4: n=3, papers=1, dirs={'0': 1, '-1': 2}; Sox2: n=3, papers=1, dirs={'0': 3}; NANOG: n=3, papers=1, dirs={'0': 3}; c-Myc: n=3, papers=1, dirs={'0': 2, '-1': 1}; p19ARF: n=3, papers=1, dirs={'0': 1, '-1': 2}; p21: n=3, papers=1, dirs={'0': 2, '-1': 1}; p53: n=3, papers=1, dirs={'0': 2, '-1': 1}; Bcl-2: n=3, papers=1, dirs={'0': 1, '-1': 1}	none detected
Proteomic	10 conditions, 3 papers; mentioned 0	DNA damage marker (4), Apoptosis marker (4), Secreted Tumor Biomarkers (3), Autophagy (3), MAPK signaling (3)	gamma-H2AX: n=7, papers=2, dirs={'0': 7}; cleaved PARP: n=4, papers=1, dirs={'0': 4}; DKK1: n=3, papers=1, dirs={'-1': 2, '0': 1}; SFRP1: n=3, papers=1, dirs={'0': 2, '-1': 1}; Beclin-1: n=3, papers=1, dirs={'1': 1, '-1': 1, '0': 1}; LC3-I: n=3, papers=1, dirs={'0': 1, '1': 2}; p-ERK1/2: n=3, papers=1, dirs={'0': 3}; ERK1/2: n=3, papers=1, dirs={'0': 3}	none detected

4.6.6 Mechanism-of-action synthesis within subgroup

Table 4.6g. Parameter combinations with multi-endpoint coherence in RF/mobile-frequency electromagnetic exposures.

Condition ID	Desired-outcome count	Exposure	Coherent outcome pattern
RF-03	7	LTE (UHF-EMF); LTE (WCDMA modulation); 1700000000 Hz; SAR 2 W/kg	ASC Apoptosis 0 not reported%; Huh7 Proliferation -1 27%; Huh7 Senescence 1 not reported%; Huh7 ROS 1 not reported%; Hep3B Proliferation -1 36%; HeLa Proliferation -1 28%
RF-02	7	LTE (UHF-EMF); LTE (WCDMA modulation); 1700000000 Hz; SAR 1 W/kg	ASC Proliferation 0 88%; ASC Apoptosis 0 not reported%; Huh7 Proliferation -1 79%; Huh7 ROS 1 not reported%; Hep3B Proliferation -1 64%; HeLa Proliferation -1 66%
RF-01	5	REAC (RF-EMF)	MCF-7 Viability -1 95%; MCF-7 Senescence 1 116.6%; MCF-7 Viability -1 96.7%; MCF-7 Viability -1 84.53%; MCF-7 Senescence 1 133%
RF-08	3	RF-EMF (MF-EMF); Amplitude-modulated sine; 13560000 Hz	HT29 Apoptosis 1 134%; SW480 Apoptosis 1 145%; LoVo Apoptosis 1 130%
RF-09	2	RF-EMF (MF-EMF); Continuous sine; 13560000 Hz	SW480 Apoptosis 1 145%; LoVo Apoptosis 1 130%

Interpretation: within this subgroup, the strongest monotherapy signal should be assigned to conditions where cancer-cell fate and mechanistic stress move together while healthy rows remain directionally neutral. Conditions with strong cancer effects but absent healthy controls are hypothesis-generating rather than selective evidence.

4.6.7 Limitations of subgroup evidence base

The category is physically heterogeneous and often lacks matched healthy tissue controls. SAR-based reductions in cancer proliferation are not sufficient for monotherapy selectivity when healthy cells are also suppressed.

4.7. Millimeter-wave/terahertz radiative exposures

4.7.1 Subgroup overview

This category contains high-frequency radiative exposures with power density as the primary dose metric. It offers one of the strongest matched selectivity signals in H1299 versus MCF-10A under W-band MMW, plus melanoma/prostate/lymphoid THz observations. The retained evidence base is 40 conditions from 4 papers, with a cancer-to-healthy row ratio of 24:16.

Per-study cancer/healthy ratio: P11:6C/0H; P12:7C/15H; P32:4C/0H; P33:7C/1H.

Direct selectivity is possible in papers with both cancer and healthy/non-cancer lines: P12, P33.

Dominant tested biological contexts: Breast epithelial cells (15 rows); Non-small cell lung cancer (NSCLC) (13 rows); Melanotic melanoma (4 rows); Burkitt lymphoma (B-cell origin, EBV-negative, homozygous mutated TP53) (2 rows); T-cell acute lymphoblastic leukemia (2 rows).

Table 4.7a. Cell-line panel for Millimeter-wave/terahertz radiative exposures.

Role	Tumour/tissue type	Species/origin	Cell lines (number of conditions)	Rows
Cancer	Non-small cell lung cancer (NSCLC)	Human	H1299 (13)	13
Cancer	Melanotic melanoma	Human	A375 (4)	4
Cancer	Burkitt lymphoma (B-cell origin, EBV-negative, homozygous mutated TP53)	Human	Ramos (2)	2
Cancer	Invasive breast adenocarcinoma (wild-type TP53)	Human	MCF-7 (2)	2
Cancer	T-cell acute lymphoblastic leukemia	Human	Jurkat (2)	2
Cancer	Prostate carcinoma (wild-type p53)	Human	LNCaP (1)	1
Healthy/non-cancer	Breast epithelial cells	Human	MCF-10A (15)	15
Healthy/non-cancer	Primary mononuclear blood cells (monocytes, B-lymphocytes, T-lymphocytes, NK cells)	Human	PBMCs (1)	1

4.7.2 Exposure parameters and rationale

The subgroup contains 4 distinct physical exposure signatures after collapsing rows with identical field parameters. Full condition IDs are provided in Appendix A and are used throughout this module.

Table 4.7b. Parameter envelope for Millimeter-wave/terahertz radiative exposures.

Parameter	Values present in retained non-thermal rows
Field labels	MMW (EHF-EMF); THz radiation (EHF-EMF)
Waveforms	Sine regime (frequency skipping across W-band 75-110 GHz) - CW; Sine regime (frequency skipping across W-band 75-105 GHz) - CW; Continuous sine; Continuous
AC frequencies	numeric envelope -110000000000-140000000000; 75000000000-110000000000 Hz; 75000000000-105000000000 Hz; 35200000000 Hz; 140000000000 Hz
Power density	numeric envelope 0.16-7.4; 0.2 mW/cm ² ; 0.16 mW/cm ² ; 7.4 mW/cm ²
Devices	Scanning 8360B Series Swept Signal Generator with 8757D Scalar Network Analyzer (Agilent), solid-state frequency multiplier (x6), and SGH-10 pyramidal horn antenna (Millitech Inc.); KFA-100A MMW therapeutic instrument (Zhongcheng Kangfu Technology); IMPATT (impact ionization avalanche transit-time) diode-based radiation source with rectangular horn antenna (14 mm x 8 mm) connected via WR-6 waveguide

Frequency rationale categories: W-band anticancer potential (22); IEEE safety limit (8); W-band sweeping regime (MMW range) (6); Therapeutic MMW frequency (4).

Intensity rationale categories: Low non-thermal power (22); IEEE safety limit (8); Non-thermal power density (6); Simulation-optimized dose (4).

Most frequent frequency-rationale details: 22 rows: The W-band (75-105 GHz) millimeter wave frequency range was selected because torsional Eigen modes of biomolecules such as carbohydrates, proteins, and DNA have estimated

Eigen frequencies around 70 GHz, and a sweeping ... | 8 rows: The authors chose 0.14 THz (140 GHz) considering the upper limit of THz radiation exposure according to the IEEE standards for safety levels with respect to human exposure to radiofrequency electromagnetic fields, where... | 6 rows: The W-band (75-110 GHz) was selected to investigate non-thermal effects of millimeter waves on cancer cells, building on prior group findings of MMW effects on human lung cancer cells in this frequency range.

Most frequent intensity-rationale details: 22 rows: Power density was maintained constant at 0.2 mW/cm² (2 W/m²) to ensure non-thermal biological effects on cells, consistent with previous studies showing anticancer effects at low power densities without significantly ... | 8 rows: The radiation intensity was approximately 7,4 mW/cm² (calculated as $P/(k \cdot S) = 10/(1.2 \cdot 1.12) \text{ mW/cm}^2$), chosen to be within the IEEE safety standard upper limit of 10 mW/cm² for human exposure to radiofrequency electromagn... | 6 rows: Power density of 0.2 mW/cm² was used, well below the maximum permissible exposure of 1 mW/cm² stipulated by the US Federal Communications Commission, ensuring non-thermal irradiation conditions with a predicted temper...

4.7.3 Cell fate outcomes

The table follows the requested order: viability, proliferation, apoptosis, necrosis, ferroptosis, senescence, and motility. Desired directions are interpreted separately for cancer and healthy rows; counterproductive rows are cancer rows with pro-survival, pro-proliferative, or pro-migratory direction.

Table 4.7c. Cell-fate outcomes for Millimeter-wave/terahertz radiative exposures.

Outcome	Measured rows	Direction distribution	Strongest desirable cancer effect	Cancer counter-signal	Matched selectivity cases
Viability	11 cancer / 1 healthy (2 papers)	Cancer -1:10; 0:1 Healthy -1:1	EHF-03 / P32 / A375: 36.7% (assay CCK-8, exposure 1.5 h, observation 4 h, value 36.7%); MMW (EHF-EMF); Continuous sine; 35200000000 Hz; PD 0.16 mW/cm ²	none	mixed selectivity: 1
Proliferation	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Apoptosis	11 cancer / 15 healthy (2 papers)	Cancer 0:1; +1:10 Healthy 0:15	EHF-02 / P12 / H1299: 1542.1% (assay DAPI staining (cell death/mortality), exposure 0.0333 h, observation 168 h, value 1542.1%); MMW (EHF-EMF); Sine regime (frequency skipping across W-band 75-105 GHz) - CW; 75000000000-105000000000 Hz; PD 0.2 mW/cm ²	none	good selectivity: 6; negligible/unclear selectivity: 1
Necrosis	4 cancer / 0 healthy (1 papers)	Cancer 0:4 Healthy not measured	none	none	none
Ferroptosis	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none
Senescence	7 cancer / 15 healthy (1 papers)	Cancer 0:1; +1:6 Healthy 0:15	EHF-02 / P12 / H1299: 430.5% (assay Beta-galactosidase (SA-beta-Gal) colorimetric assay, exposure 0.0667 h, observation 336 h, value 430.5%); MMW (EHF-EMF); Sine regime (frequency skipping across W-band 75-105 GHz) - CW; 75000000000-105000000000 Hz; PD 0.2 mW/cm ²	none	good selectivity: 6; negligible/unclear selectivity: 1
Motility	0 cancer / 0 healthy (0 papers)	Cancer not measured Healthy not measured	none	none	none

Table 4.7d. Matched good-selectivity examples for Millimeter-wave/terahertz radiative exposures.

Outcome	Condition ID	Outcome exposure/observation	Cancer rows	Healthy rows
Apoptosis	EHF-02	exp/obs 0.0333/1.5 h	H1299 1 356.4%	MCF-10A 0 73.5%
Apoptosis	EHF-02	exp/obs 0.0333/168 h	H1299 1 1542.1%	MCF-10A 0 120%
Apoptosis	EHF-02	exp/obs 0.0667/1.5 h	H1299 1 366.7%	MCF-10A 0 153.1%
Apoptosis	EHF-02	exp/obs 0.0667/168 h	H1299 1 436.8%	MCF-10A 0 220%
Apoptosis	EHF-02	exp/obs 0.0667/336 h	H1299 1 332.3%	MCF-10A 0 143.2%
Apoptosis	EHF-02	exp/obs 0.1667/1.5 h	H1299 1 1200%	MCF-10A 0 108.2%
Senescence	EHF-02	exp/obs 0.0333/1.5 h	H1299 1 182.8%	MCF-10A 0 221.4%
Senescence	EHF-02	exp/obs 0.0333/336 h	H1299 1 159.7%	MCF-10A 0 72.1%
Senescence	EHF-02	exp/obs 0.0667/1.5 h	H1299 1 198.4%	MCF-10A 0 242.9%

Outcome	Condition ID	Outcome exposure/observation	Cancer rows	Healthy rows
Senescence	EHF-02	exp/obs 0.0667/168 h	H1299 1 176.5%	MCF-10A 0 95%
Senescence	EHF-02	exp/obs 0.0667/336 h	H1299 1 430.5%	MCF-10A 0 116.3%
Senescence	EHF-02	exp/obs 0.1667/1.5 h	H1299 1 323.1%	MCF-10A 0 278.6%

Dose/time and risk signals.

W-band MMW at 0.2 mW/cm² shows a delayed lethal/senescent trajectory in H1299; 2-4 min exposure produced substantial death/senescence at later time points while MCF-10A remained directionally non-apoptotic. THz at 140 GHz was less selective because PBMC viability was also reduced.

4.7.4 Mechanistic outcomes

Table 4.7e. Mechanistic coverage and fate linkage for Millimeter-wave/terahertz radiative exposures.

Mechanism	Coverage	Observed direction/effect	Same-row links to fate outcomes
ROS	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Cancer not measured Healthy not measured	no same-row fate linkage
Antioxidant	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Cancer not measured Healthy not measured	no same-row fate linkage
Membrane Potential	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Cancer not measured Healthy not measured	no same-row fate linkage
Calcium	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Cellular Morphology	13 cancer / 16 healthy (3 papers); mentioned 0 (0 papers)	Measured positive/effect: 14; no effect: 15	EHF-02 H1299: Apoptosis 1 (356.4%), Senescence 1 (182.8%); EHF-02 H1299: Apoptosis 1 (1542.1%), Senescence 0 (119.1%); EHF-02 H1299: Apoptosis 0 (109.7%), Senescence 1 (159.7%)
Cytoskeleton	0 cancer / 0 healthy (0 papers); mentioned 6 (1 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Metabolic Reprogramming	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Differentiation	0 cancer / 0 healthy (0 papers); mentioned 0 (0 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage
Cell Cycle	0 cancer / 0 healthy (0 papers); mentioned 13 (2 papers)	Measured positive/effect: 0; no effect: 0	no same-row fate linkage

W-band MMW at 0.2 mW/cm² produces delayed H1299 death/senescence with little direct MCF-10A death direction under matched conditions, suggesting a stress-threshold or repair-capacity difference rather than immediate thermal toxicity. 35.2 GHz A375 exposure links viability loss to caspase-3/8 activation, while THz effects appear less selective because PBMC viability is also reduced.

4.7.5 Molecular outcomes

Table 4.7f. Genomic and proteomic findings for Millimeter-wave/terahertz radiative exposures.

Block	Coverage	Frequently studied families	Recurrent targets and direction consistency	Cancer/healthy opposite-direction signatures
Genomic	not investigated	none	none	none
Proteomic	4 conditions, 1 papers; mentioned 0	Caspases (4)	Cleaved caspase-3: n=4, papers=1, dirs=['1': 4]; Cleaved caspase-8: n=4, papers=1, dirs=['1': 4]	none detected

4.7.6 Mechanism-of-action synthesis within subgroup

Table 4.7g. Parameter combinations with multi-endpoint coherence in Millimeter-wave/terahertz radiative exposures.

Condition ID	Desired-outcome count	Exposure	Coherent outcome pattern
EHF-02	42	MMW (EHF-EMF); Sine regime (frequency skipping across W-band 75-105 GHz) - CW; 75000000000-105000000000 Hz; PD 0.2 mW/cm ²	H1299 Apoptosis 1 356.4%; H1299 Senescence 1 182.8%; H1299 Apoptosis 1 1542.1%; H1299 Senescence 1 159.7%; H1299 Apoptosis 1 366.7%; H1299 Senescence 1 198.4%
EHF-03	8	MMW (EHF-EMF); Continuous sine; 35200000000 Hz; PD 0.16 mW/cm ²	A375 Viability -1 76%; A375 Apoptosis 1 214.5%; A375 Viability -1 73%; A375 Apoptosis 1 224.6%; A375 Viability -1 62%; A375 Apoptosis 1 288.7%
EHF-04	6	THz radiation (EHF-EMF); Continuous; 140000000000 Hz; PD 7.4 mW/cm ²	Ramos Viability -1 75%; Ramos Viability -1 73%; Jurkat Viability -1 75%; Jurkat Viability -1 73%; MCF-7 Viability -1 90%; LNCaP Viability -1 63%

Interpretation: within this subgroup, the strongest monotherapy signal should be assigned to conditions where cancer-cell fate and mechanistic stress move together while healthy rows remain directionally neutral. Conditions with strong cancer effects but absent healthy controls are hypothesis-generating rather than selective evidence.

4.7.7 Limitations of subgroup evidence base

Coverage is driven by few papers. Mechanistic molecular data are mostly caspases in one A375 study, and direct ROS/metabolic profiling is absent. MMW/THz dosimetry is sensitive to geometry, depth, hydration, and exposure surface, limiting direct transfer from monolayer to tissue.

5. Cross-subgroup synthesis

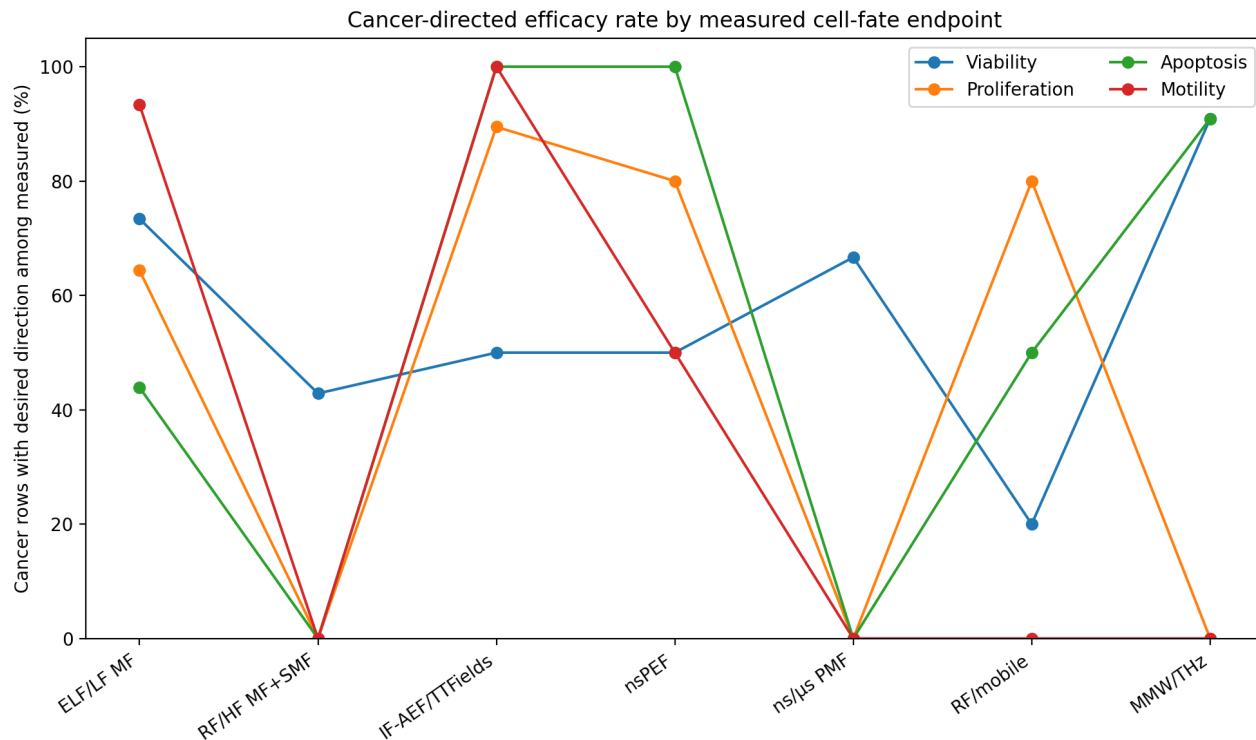


Figure 3. Cancer-directed efficacy rates among measured rows for selected fate endpoints. Absence of a bar/line value may reflect no measurement, not absence of effect.

5.1 Shared mechanistic signatures

- Stress-threshold selectivity rather than simple field toxicity. The best windows generally do not damage every cell equally: TTFields suppress MDA-MB-231 migration/proliferation while MCF-12 is preserved; nsPEF elevates ROS in MCF-7 while MCF-10A ROS decreases; W-band MMW kills/senesces H1299 without a matching MCF-10A apoptosis direction.
- Mitochondrial/ROS involvement recurs across physically distinct subgroups. ELF/LF fields, RF/HF magnetic hybrids, nsPEF, and AEF all contain ROS-positive cancer rows, but only nsPEF breast comparison and selected ELF/LF glioma comparisons show clear healthy sparing of ROS.
- Cell-cycle/mitotic vulnerability is a cross-cutting theme. TTFields are explicitly anti-mitotic; ELF/LF MEMF/PEMF rows show cell-cycle changes and senescence; MMW rows show delayed senescence; RF/REAC rows show p21/p53/senescence markers in limited contexts.
- Morphological remodeling is frequent but not sufficient as a selectivity biomarker. It is well-covered in nsPEF, MMW/THz, and ELF/LF categories, yet the causal chain is strongest only when morphology co-occurs with viability/proliferation/apoptosis outcomes.

5.2 Field-specific mechanisms

Table 5. Field-specific mechanistic signatures.

Macro-category	Candidate field-specific mechanism
ELF/LF magnetic-field platforms	Windowed low-frequency redox/cell-cycle effects, senescence, apoptosis/necrosis at specific high-intensity or pulsed settings; motility inhibition in renal carcinoma under 50 Hz/4.5 mT.
RF/HF magnetic field + static-field hybrids	Mitochondrial Fe-S/hyperfine-resonance hypothesis with narrow frequency windows; currently not selective due absent healthy controls and pro-growth windows.

Macro-category	Candidate field-specific mechanism
Intermediate-frequency alternating electric fields	Mitotic spindle/cytokinesis disruption and motility suppression; strongest matched evidence in TNBC/normal breast epithelial comparisons.
nsPEF	Nanosecond electroporation/electroperturbation of membranes and organelles; ATP leakage, mitochondrial potential collapse, ROS, apoptosis.
Ultrashort pulsed magnetic fields	Pulse waveform and dB/dt window effect in C6 glioma; insufficient mechanistic breadth.
RF/mobile	SAR/device-dependent proliferation and senescence effects; selectivity weak because healthy cells can also be suppressed.
MMW/THz	High-frequency radiative stress leading to delayed apoptosis/senescence and caspase activation; geometry/dosimetry likely critical.

5.3 Tumour-type response patterns

Breast cancer is the most cross-category responsive tissue context. MDA-MB-231 responds to TTFields/AEF with strong viability/motility suppression and to selected ELF/LF magnetic fields with senescence/proliferation effects, while MCF-7 responds to nsPEF and some ELF/LF or RF/REAC conditions. Glioma/glioblastoma models respond strongly to nsPEF spheroid pulses and selected ELF/LF/MEMP conditions, but healthy neural comparator coverage varies. Lung cancer shows strong MMW selectivity in H1299 versus MCF-10A and weaker/more heterogeneous ELF/LF/RF responses. Melanoma responses are configuration-specific: B16F10 responds to AEF/ELF conditions and A375 responds to 35.2 GHz MMW or nsPEF, but healthy-melanocyte comparators are absent.

5.4 Testable experimental hypotheses for a future project

Table 6. Proposed experimental hypotheses emerging from cross-subgroup synthesis.

Hypothesis	Field configuration	Target cell lines	Primary outcomes	Expected selectivity window	Mechanistic rationale
H1: TTFields/AEF TNBC selectivity	150 kHz alternating electric field at 1.5 V/cm; compare uniform versus frequency-modulated mode; 24-48 h exposure.	MDA-MB-231 and HCC38 versus MCF-12/MCF-10A.	Primary: motility and proliferation; secondary: apoptosis and mitotic spindle/cytoskeleton imaging.	Expected window: MDA-MB-231 motility 19.6-30.4% of control with MCF-12 motility 93-100%; viability suppression at 24-48 h with healthier preservation than at 72 h.	TTFields have the cleanest matched selectivity pattern and a plausible mitotic/cytoskeletal mechanism; FM may improve anti-migratory effect.
H2: nsPEF oxidative-organelle threshold in breast cancer	50 ns nsPEF, 20 kV/cm, 2 kHz, short exposure (2-5 min equivalent); pulse number standardized across viability, ROS, mitochondrial potential, and apoptosis assays.	MCF-7 and MDA-MB-231 versus MCF-10A.	Primary: ROS and apoptosis; secondary: mitochondrial membrane potential, ATP leakage versus clonogenic survival.	Expected window: cancer ROS increase (~150% in MCF-7) with MCF-10A ROS decrease (~50%) and cancer viability/apoptosis directionally affected while MCF-10A remains neutral.	The matched MCF-7/MCF-10A rows provide the clearest non-thermal oxidative selectivity signal and directly connect ROS with cell fate.
H3: ELF-PEMF senescence/proliferation window	8 Hz full-square ELF-PEMF at 0.011 T for 5 h.	MDA-MB-231 and MCF-7 versus FF95/MCF-10A; add tissue-matched breast epithelial controls.	Primary: senescence (GL13/SenTraGor, SA-beta-Gal) and proliferation; secondary: p53/p21/CDK2/cyclin A and DNA damage.	Expected window: MDA-MB-231 senescence 952% and proliferation 16%; MCF-7 senescence 400% and proliferation 33%; FF95 senescence 100% and proliferation 123%.	This is a strong phenotype but currently anchored to a narrow evidence base; replication with matched breast controls would test whether the senescence window is cancer-selective.
H4: W-band MMW delayed death/senescence in NSCLC	75-105 GHz frequency-skipping continuous W-band MMW at 0.2 mW/cm ² for 2-4 min.	H1299 versus MCF-10A and bronchial epithelial controls.	Primary: apoptosis/death at day 1/day 7 and senescence at day 7/day 14; secondary: caspase, p53/p21, DNA damage, ROS.	Expected window: H1299 apoptosis/death 356-1542% versus MCF-10A directionally non-apoptotic in matched rows; delayed senescence selectively accumulates in surviving cancer cells.	This is the strongest radiative high-frequency selectivity signal, but needs mechanistic expansion beyond morphology/death and more epithelial controls.
H5: RF/HF magnetic resonance as a redox screen, not primary monotherapy	2.0-5.0 MHz RF-MF with 45 uT SMF and nT-uT AC amplitude; fine frequency steps around 3.5-4.5 MHz.	HT-1080 plus normal fibroblasts and mitochondrial-rescue/inhibitor controls.	Primary: mitochondrial superoxide and oxygen consumption; secondary: proliferation/viability.	Expected window: ROS-positive, mildly inhibitory 2-3.5 MHz conditions contrasted with pro-growth 4-4.5 MHz conditions.	The narrow reversal from ROS/inhibition to pro-growth is mechanistically valuable and a safety screen for resonance-based field selection.

6. Global limitations

- Coverage imbalance. ELF/LF magnetic-field platforms contribute 222 rows and 27 papers, while RF/HF hybrids have 21 rows from two papers and ultrashort pulsed magnetic fields have 24 rows from one paper.
- Outcome inconsistency. Viability is common, but ferroptosis, differentiation, antioxidant response, calcium, and direct cell-cycle assays are sparse in many field categories.
- Study-level collapse. Some high coverage percentages are caused by prolific single studies; this is especially clear for ultrashort pulsed magnetic fields, MMW H1299/MCF-10A, and the nsPEF spheroid pulse series.
- Healthy comparator mismatch. Several apparently strong cancer-killing conditions lack a paired healthy line; when healthy lines exist, they are often not tissue-matched, and some are transformed or non-primary.
- Dosimetry heterogeneity. Field labels often obscure major physical differences: magnetic versus electric dominance, SAR versus power density, static bias, pulse width, duty cycle, and device geometry. This prevents simple pooling.
- Direction versus magnitude tension. Direction encodes paper-defined significance, but some Direction=0 healthy rows have large numerical shifts; therefore healthy sparing must be validated with power, effect size, and functional assays.
- Monotherapy evidence remains in vitro. The dataset supports cell-autonomous and co-culture hypotheses, but not pharmacokinetics, tissue heating at depth, immune-system contribution, or tumour microenvironment transport constraints.

7. Conclusions

Strictly within the non-thermal dataset, EMF configurations can produce antitumor monotherapy-like effects in vitro, but selectivity is configuration-specific rather than a generic property of EMF exposure. The most credible selective windows are intermediate-frequency TFields/AEF in breast cancer migration/proliferation, nsPEF oxidative-organelle stress in breast cells, selected ELF-PEMF senescence/proliferation windows, and W-band MMW delayed death/senescence in NSCLC. The strongest mechanistic convergence is not thermal injury; it is field-specific amplification of pre-existing cancer vulnerabilities: mitotic instability, redox imbalance, membrane/organelle fragility, and impaired stress recovery. Future studies should use matched cancer/normal panels, harmonized exposure IDs, simultaneous fate and mechanism measurements, and blinded thermal monitoring. A future project should prioritize TFields/AEF and nsPEF as the most immediately testable selective monotherapy candidates, while using ELF/PEMF and W-band MMW as high-upside mechanisms requiring stronger independent replication.

Appendix A. Full exposure-condition lookup tables

Each condition ID represents one unique combination of physical exposure parameters after thermal filtering. Rows with identical exposure parameters but different cell lines/outcomes are collapsed into the same condition ID. These IDs are referenced throughout the report.

Appendix A.1. ELF/LF magnetic-field platforms

Appendix Table A.1. Exposure lookup for ELF/LF magnetic-field platforms.

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
LF-01	P1	3	ELF-EMF	Square pulse	20 Hz	100 μ T	N/A N/A	E N/A N/A; PD N/A N/A		CT2A(1), N2A(1), C8D1A(1)	RILZ coil system (custom design, CTB – Universidad Politécnica de Madrid)
LF-02	P1	3	ELF-EMF	Square pulse	30 Hz	100 μ T	N/A N/A	E N/A N/A; PD N/A N/A		CT2A(1), N2A(1), C8D1A(1)	RILZ coil system (custom design, CTB – Universidad Politécnica de Madrid)
LF-03	P1	2	ELF-EMF	Square pulse	40 Hz	100 μ T	N/A N/A	E N/A N/A; PD N/A N/A		N2A(1), C8D1A(1)	RILZ coil system (custom design, CTB – Universidad Politécnica de Madrid)
LF-04	P1	6	ELF-EMF	Square pulse	45 Hz	100 μ T	N/A N/A	E N/A N/A; PD N/A N/A		CT2A(2), N2A(2), C8D1A(2)	RILZ coil system (custom design, CTB – Universidad Politécnica de Madrid)
LF-05	P1	5	ELF-EMF	Square pulse	50 Hz	100 μ T	N/A N/A	E N/A N/A; PD N/A N/A		N2A(2), C8D1A(2), CT2A(1)	RILZ coil system (custom design, CTB – Universidad Politécnica de Madrid)
LF-06	P1	6	ELF-EMF	Square pulse	55 Hz	100 μ T	N/A N/A	E N/A N/A; PD N/A N/A		CT2A(2), N2A(2), C8D1A(2)	RILZ coil system (custom design, CTB – Universidad Politécnica de Madrid)
LF-07	P1	1	ELF-EMF	Square pulse	60 Hz	100 μ T	N/A N/A	E N/A N/A; PD N/A N/A		CT2A(1)	RILZ coil system (custom design, CTB – Universidad Politécnica de Madrid)
LF-08	P2	2	ELF-EMF	Sine	10 Hz	70 μ T	45 μ T	E N/A N/A; PD N/A N/A		PANC-1(2)	Coaxial coils
LF-09	P2	2	ELF-EMF	Sine	14 Hz	70 μ T	45 μ T	E N/A N/A; PD N/A N/A		PANC-1(2)	Coaxial coils
LF-10	P2	2	ELF-EMF	Sine	3 Hz	70 μ T	45 μ T	E N/A N/A; PD N/A N/A		PANC-1(2)	Coaxial coils
LF-11	P2	2	ELF-EMF	Sine	4 Hz	70 μ T	45 μ T	E N/A N/A; PD N/A N/A		1627(2)	Coaxial coils
LF-12	P2	7	ELF-EMF	Sine	6 Hz	70 μ T	45 μ T	E N/A N/A; PD N/A N/A		PANC-1(2), 1627(2), MCF-7(2), SKBR3(1)	Coaxial coils
LF-13	P2	2	ELF-EMF	Sine	8 Hz	70 μ T	45 μ T	E N/A N/A; PD N/A N/A		MCF-7(1), SKBR3(1)	Coaxial coils
LF-14	P3	1	ELF P-EMF (ELF-EMF)	Sine	50 Hz	22.6 mT	N/A N/A	E N/A N/A; PD N/A N/A		DU-145(1)	Solenoid coil (6000 turns)
LF-15	P3	1	ELF P-EMF (ELF-EMF)	Sine	50 Hz	35 mT	N/A N/A	E N/A N/A; PD N/A N/A		DU-145(1)	Solenoid coil (6000 turns)
LF-16	P4	2	AMF (LF-EMF)	Sine	160000 Hz	18 mT	N/A N/A	E N/A N/A; PD N/A N/A		U87(1), LN229(1)	Transistor inverter (Hot Shot) + solenoid copper coil
LF-17	P4	2	AMF (LF-EMF)	Sine	198000 Hz	18 mT	N/A N/A	E N/A N/A; PD N/A N/A		U87(1), LN229(1)	Transistor inverter (Hot Shot) + solenoid copper coil
LF-18	P4	12	AMF (LF-EMF)	Sine	227000 Hz	18 mT	N/A N/A	E N/A N/A; PD N/A N/A		U87(1), LN229(1), U251(1), T98(1), A172(1), PANC-1(1), MCF-7(1), MDA-MB-231(1); +more	Transistor inverter (Hot Shot) + solenoid copper coil

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
LF-19	P4	2	AMF (LF-EMF)	Sine	280000 Hz	18 mT	N/A N/A	E N/A N/A; PD N/A N/A		U87(1), LN229(1)	Transistor inverter (Hot Shot) + solenoid copper coil
LF-20	P4	2	AMF (LF-EMF)	Sine	320000 Hz	18 mT	N/A N/A	E N/A N/A; PD N/A N/A		U87(1), LN229(1)	Transistor inverter (Hot Shot) + solenoid copper coil
LF-21	P5	18	MEMP (ELF-EMF)	Pulsed (multifrequency)	<30 Hz	>2 T	N/A N/A	E N/A N/A; PD N/A N/A		U373 MG(3), A9(3), HeLa(2), MC-38-Luc(2), HaCaT(2), CV-1(2), COS-1(2), PAM(2)	Modulator equipment by Paso Alto Biotechnology Inc.
LF-22	P6	5	ELF-EMF	Not reported	1 Hz	100 mT	N/A N/A	E N/A N/A; PD N/A N/A		U87(1), U251(1), U251-P53(1), Normal human fibroblast(1), MCF-7(1)	ELF-EMF exposure device (Parto Farazane Farda, Iran) connected to a GWInstek SFG-1000 series waveform generator
LF-23	P8	2	Oscillating Magnetic Field - Helmholtz coil, fixed axis (EL...	Sine	137 Hz	5 mT	N/A N/A	E N/A N/A; PD N/A N/A		GBM115(1), DIPG(1)	Helmholtz coil driven by function generator (Wavetek) and high current amplifier (Taidacent)
LF-24	P8	4	Spinning Oscillating Magnetic Field (ELF-MF)	Sine	137 Hz	5 mT	N/A N/A	E N/A N/A; PD N/A N/A		GBM115(2), DIPG(2)	Oncomagnetic device (oncoscillator with spinning neodymium N52 permanent magnet). 1Dsp.
LF-25	P8	2	Spinning Oscillating Magnetic Field (ELF-MF)	Sine	272 Hz	5 mT	N/A N/A	E N/A N/A; PD N/A N/A		GBM115(1), DIPG(1)	Oncomagnetic device (oncoscillator with spinning neodymium N52 permanent magnet). 1Dsp.
LF-26	P9	3	ELF-EMF + SMF	Sine	50 Hz	1,0-2,5 mT	3,0-4,0 mT	E N/A N/A; PD N/A N/A		G401(1), A549(1), HEK293(1)	Self-constructed MF exposure facility based on a prototype by Tofani et al., consisting of a pair of horizontally mount...
LF-27	P10	3	ELF-PEMF (ELF-EMF)	Full square (pulsed)	8 Hz	0.011 T	N/A N/A	E N/A N/A; PD N/A N/A		MDA-MB-231(1), MCF-7(1), FF95(1)	Alphatron 4100-MW
LF-28	P13	3	ELF-EMF	Sine	0.01 Hz	100 mT	N/A N/A	E N/A N/A; PD N/A N/A		DU-145(1), HUVEC(1), K562(1)	Custom-built ELF-EMF generator with iron core magnetic circuit (two 500-turn coils in series, 1000 turns total), capaci...
LF-29	P13	1	ELF-EMF	Sine	1 Hz	100 mT	N/A N/A	E N/A N/A; PD N/A N/A		MDA-MB-231(1)	Custom-built ELF-EMF generator with iron core magnetic circuit (two 500-turn coils in series, 1000 turns total), capaci...
LF-30	P14	1	ELF-EMF	Square	30 Hz	100 μ T	N/A N/A	E N/A N/A; PD N/A N/A		PC12(1)	RILZ coils (custom-designed capsule-shaped coil system with two coils of 222 turns of enameled copper each, powered by ...
LF-31	P14	5	ELF-EMF	Square	45 Hz	100 μ T	N/A N/A	E N/A N/A; PD N/A N/A		PC12(1), B16F10(1), SKBR3(1), MDA-MB-231(1), 3T3(1)	RILZ coils (custom-designed capsule-shaped coil system with two coils of 222 turns of enameled copper each, powered by ...
LF-32	P14	5	ELF-EMF	Square	50 Hz	100 μ T	N/A N/A	E N/A N/A; PD N/A N/A		PC12(1), B16F10(1), SKBR3(1), MDA-MB-231(1), 3T3(1)	RILZ coils (custom-designed capsule-shaped coil system with two coils of 222 turns of enameled copper each, powered by ...

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
LF-33	P14	5	ELF-EMF	Square	55 Hz	100 μ T	N/A N/A	E N/A N/A; PD N/A N/A		PC12(1), B16F10(1), SKBR3(1), MDA-MB-231(1), 3T3(1)	RILZ coils (custom-designed capsule-shaped coil system with two coils of 222 turns of enameled copper each, powered by ...
LF-34	P14	1	ELF-EMF	Square	80 Hz	100 μ T	N/A N/A	E N/A N/A; PD N/A N/A		PC12(1)	RILZ coils (custom-designed capsule-shaped coil system with two coils of 222 turns of enameled copper each, powered by ...
LF-35	P15	1	ELF-PEMF (ELF-EMF)	Not reported	30 Hz	1.5-16 mT	N/A N/A	E N/A N/A; PD N/A N/A		HT-1197(1)	Helmholtz coil (11-inch diameter, 500 turns of 26-gauge copper wire) connected to a frequency generator, verified by an...
LF-36	P16	12	ELF-EMF	Sine-like (not explicitly specified)	20 Hz	5 mT	N/A N/A	E N/A N/A; PD N/A N/A		Raji(6), 293T(2), HepG2(2), A549(2)	Self-made magnetic field generator (enameled copper wire in ABS plastic shell, 450x230x25 mm)
LF-37	P18	2	ELF-EMF + SMF	Sine	14.5 Hz	10 μ T	45 μ T	E N/A N/A; PD N/A N/A		HT-1080(1), HDF (Human Dermal Fibroblast)(1)	Square Helmholtz coil inside Mu-metal shielded box, driven by FY6900 DDS waveform generator (Kkmoon) and HP 33120A 15 M...
LF-38	P18	1	ELF-EMF + SMF	Sine	15 Hz	10 μ T	45 μ T	E N/A N/A; PD N/A N/A		HDF (Human Dermal Fibroblast)(1)	Square Helmholtz coil inside Mu-metal shielded box, driven by FY6900 DDS waveform generator (Kkmoon) and HP 33120A 15 M...
LF-39	P18	1	ELF-EMF + SMF	Sine	15.5 Hz	10 μ T	45 μ T	E N/A N/A; PD N/A N/A		HT-1080(1)	Square Helmholtz coil inside Mu-metal shielded box, driven by FY6900 DDS waveform generator (Kkmoon) and HP 33120A 15 M...
LF-40	P18	1	ELF-EMF + SMF	Sine	16.5 Hz	18 μ T	45 μ T	E N/A N/A; PD N/A N/A		HT-1080(1)	Square Helmholtz coil inside Mu-metal shielded box, driven by FY6900 DDS waveform generator (Kkmoon) and HP 33120A 15 M...
LF-41	P18	1	ELF-EMF + SMF	Sine	16.5 Hz	18.5 μ T	45 μ T	E N/A N/A; PD N/A N/A		HT-1080(1)	Square Helmholtz coil inside Mu-metal shielded box, driven by FY6900 DDS waveform generator (Kkmoon) and HP 33120A 15 M...
LF-42	P18	1	ELF-EMF + SMF	Sine	33 Hz	10 μ T	45 μ T	E N/A N/A; PD N/A N/A		HT-1080(1)	Square Helmholtz coil inside Mu-metal shielded box, driven by FY6900 DDS waveform generator (Kkmoon) and HP 33120A 15 M...
LF-43	P18	2	SMF	N/A	N/A N/A	N/A N/A	-45 μ T	E N/A N/A; PD N/A N/A		HT-1080(2)	Square Helmholtz coil inside Mu-metal shielded box, driven by FY6900 DDS waveform generator (Kkmoon) and HP 33120A 15 M...
LF-44	P19	8	ELF-MF (ELF-EMF)	Not reported	50 Hz	0.1 mT	N/A N/A	E N/A N/A; PD N/A N/A		MDA-MB-231(3), MCF-7(3), MCF-10A(2)	Square-shaped Helmholtz coil with

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
											electrical current generator
LF-45	P19	10	ELF-MF (ELF-EMF)	Not reported	50 Hz	1 mT	N/A N/A	E N/A N/A; PD N/A N/A		MCF-10A(4), MDA-MB-231(3), MCF-7(3)	Square-shaped Helmholtz coil with electrical current generator
LF-46	P20	2	SREMF (ELF-EMF)	Square	7.83 Hz	200 μ T	N/A N/A	E N/A N/A; PD N/A N/A		HeLa(1), NIH/3T3(1)	Honeycomb-like emitter array (custom-designed ring-shaped EMF emitters arranged in hexagonal pattern, Array 1, d=23 mm,...
LF-47	P20	2	SREMF (ELF-EMF)	Square	7.83 Hz	600 μ T	N/A N/A	E N/A N/A; PD N/A N/A		HeLa(1), NIH/3T3(1)	Honeycomb-like emitter array (custom-designed ring-shaped EMF emitters arranged in hexagonal pattern, Array 1, d=23 mm,...
LF-48	P21	3	ELF-EMF	Complex patterned (modeled after interc...	Not reported Not reported	1 μ T	N/A N/A	E N/A N/A; PD N/A N/A		B16-BL6(2), HEK293T(1)	30 Ohm Helmholtz coil (39 cm x 39 cm box wrapped by 305 m of 30 AWG insulated copper wire), connected to a digital-to-a...
LF-49	P21	2	ELF-EMF	Sine	3 Hz	1 μ T	N/A N/A	E N/A N/A; PD N/A N/A		B16-BL6(1), HEK293T(1)	30 Ohm Helmholtz coil (39 cm x 39 cm box wrapped by 305 m of 30 AWG insulated copper wire), connected to a digital-to-a...
LF-50	P23	1	Gradient RMF (ELF-MF)	Periodic	2.5 Hz	0.41 T	N/A N/A	E N/A N/A; PD N/A N/A		MDA-MB-231(1)	Custom gradient RMF cell culture platform
LF-51	P23	1	Gradient RMF (ELF-MF)	Periodic	5 Hz	0.04 T	N/A N/A	E N/A N/A; PD N/A N/A		MDA-MB-231(1)	Custom gradient RMF cell culture platform
LF-52	P23	1	Gradient RMF (ELF-MF)	Periodic	5 Hz	0.2 T	N/A N/A	E N/A N/A; PD N/A N/A		MDA-MB-231(1)	Custom gradient RMF cell culture platform
LF-53	P23	3	Gradient RMF (ELF-MF)	Periodic	5 Hz	0.41 T	N/A N/A	E N/A N/A; PD N/A N/A		MDA-MB-231(1), BT549(1), MDA-MB-468(1)	Custom gradient RMF cell culture platform
LF-54	P24	4	ELF-EMF	Not reported	50; 385 Hz	20 mT	N/A N/A	E N/A N/A; PD N/A N/A		A549(1), LLC(1), hLEC(1), MDA-MB-231(1)	Solenoid magnet (#R-2016-12; Magnatech) interrupted by a circuit and connected to a BioRad Power Pack (10 V DC)
LF-55	P25	1	ELF-EMF	Not reported	1 Hz	100 mT	N/A N/A	E N/A N/A; PD N/A N/A		MC4-L2(1)	Custom-built U-shape solenoid coil system with ionic core connected to a GW Instek SFG-1000 wave form generator
LF-56	P28	4	ELF-EMF	Sine	50 Hz	4.5 mT	N/A N/A	E N/A N/A; PD N/A N/A		HEK293(1), 786-O(1), 769-P(1), Caki1(1)	Helmholtz coil
LF-57	P36	1	ELF-EMF + SMF	Sine	24 Hz	100 μ T	45 μ T	E N/A N/A; PD N/A N/A		HT-29(1)	Two independent pairs of coaxial coils wound into a cylindrical frame (outer radius 8 cm, distance between coil couples...
LF-58	P36	1	ELF-EMF + SMF	Sine	6 Hz	100 μ T	45 μ T	E N/A N/A; PD N/A N/A		MDA-MB-231(1)	Two independent pairs of coaxial coils wound into a cylindrical frame (outer radius 8 cm, distance between coil couples...
LF-59	P38	8	EMF (LF-EMF)	Sine	207000 Hz	1.4 mT	N/A N/A	E N/A N/A; PD N/A N/A		HDFa(2), NHA(2), SH-SY5Y(2), T98G(2)	Wireless power transfer (WPT)

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
											system prototype with high-frequency inverter and coupling coils
LF-60	P38	8	EMF (LF-EMF)	Sine	87000 Hz	1.6 mT	N/A N/A	E N/A N/A; PD N/A N/A		HDFa(2), NHA(2), SH-SY5Y(2), T98G(2)	Wireless power transfer (WPT) system prototype with high-frequency inverter and coupling coils
LF-61	P40	2	LF-RMF (ELF-RMF)	Sine	4.2 Hz	0.1 T	N/A N/A	E N/A N/A; PD N/A N/A		MDA-MB-231(1), MCF-7(1)	Custom-built rotating magnetic field apparatus using rare-earth permanent magnets (N42M) on a rotor with opposite poles...
LF-62	P40	4	LF-RMF (ELF-RMF)	Sine	4.2 Hz	0.4 T	N/A N/A	E N/A N/A; PD N/A N/A		MDA-MB-231(1), MCF-7(1), MCF-10A(1), RPE1(1)	Custom-built rotating magnetic field apparatus using rare-earth permanent magnets (N42M) on a rotor with opposite poles...
LF-63	P41	2	ELF-EMF	Sine	50 Hz	1 mT	N/A N/A	E N/A N/A; PD N/A N/A		MCF-10A(1), MDA-MB-231(1)	Self-designed square Helmholtz coil (inner coil sides 34x34 cm, distance between solenoids 19.5 cm) used inside a cell ...
LF-64	P44	2	AMF (LF-EMF)	Sine	100000 Hz	10 mT	N/A N/A	E N/A N/A; PD N/A N/A		MDA-MB-231(2)	Leybold 562 15 coil (N=1000 turns, L=36 mH, R=9.5 Ohm, I _{max} =1.25A) connected to Leybold 522 63 power generator (0-30 Vp...
LF-65	P44	2	SMF	N/A	N/A N/A	N/A N/A	15 mT	E N/A N/A; PD N/A N/A		MDA-MB-231(2)	Leybold 562 15 coil (N=1000 turns, L=36 mH, R=9.5 Ohm, I _{max} =1.25A) connected to Leybold 522 63 power generator (0-30 Vp...
LF-66	P48	3	PEMF (ELF-EMF)	Sine	20 Hz	3 mT	N/A N/A	E N/A N/A; PD N/A N/A		HeLa-LUC(1), CaSki-LUC(1), SiHa-LUC(1)	Two Helmholtz coils connected at top and bottom of culture rack, housed within standard cell culture incubator
LF-67	P48	2	PEMF (ELF-EMF)	Sine	60 Hz	1.2 mT	N/A N/A	E N/A N/A; PD N/A N/A		HeLa-LUC (co-culture with CD8+ T cells)(1), CD8+ T cells (from CC patient tumor tissues)(1)	Two Helmholtz coils connected at top and bottom of culture rack, housed within standard cell culture incubator

Appendix A.2. RF/HF magnetic field + static-field hybrids

Appendix Table A.2. Exposure lookup for RF/HF magnetic field + static-field hybrids.

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
RFM-01	P7	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	2000000 Hz	20 nT	45 µT	E N/A N/A; PD N/A N/A		HT-1080(1)	Square Helmholtz coils (30 turns each, side=15 cm, 1.63 mm wire) within a mu-metal shielding box (MuShield Company); RF...
RFM-02	P7	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	2500000 Hz	20 nT	45 µT	E N/A N/A; PD N/A N/A		HT-1080(1)	Square Helmholtz coils (30 turns each, side=15 cm, 1.63

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
											mm wire) within a mu-metal shielding box (MuShield Company); RF...
RFM-03	P7	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	3000000 Hz	20 nT	45 µT	E N/A N/A; PD N/A N/A		HT-1080(1)	Square Helmholtz coils (30 turns each, side=15 cm, 1.63 mm wire) within a mu-metal shielding box (MuShield Company); RF...
RFM-04	P7	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	3500000 Hz	20 nT	45 µT	E N/A N/A; PD N/A N/A		HT-1080(1)	Square Helmholtz coils (30 turns each, side=15 cm, 1.63 mm wire) within a mu-metal shielding box (MuShield Company); RF...
RFM-05	P7	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	4000000 Hz	20 nT	45 µT	E N/A N/A; PD N/A N/A		HT-1080(1)	Square Helmholtz coils (30 turns each, side=15 cm, 1.63 mm wire) within a mu-metal shielding box (MuShield Company); RF...
RFM-06	P7	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	4500000 Hz	20 nT	45 µT	E N/A N/A; PD N/A N/A		HT-1080(1)	Square Helmholtz coils (30 turns each, side=15 cm, 1.63 mm wire) within a mu-metal shielding box (MuShield Company); RF...
RFM-07	P7	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	5000000 Hz	20 nT	45 µT	E N/A N/A; PD N/A N/A		HT-1080(1)	Square Helmholtz coils (30 turns each, side=15 cm, 1.63 mm wire) within a mu-metal shielding box (MuShield Company); RF...
RFM-08	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Frequency modulated at 50 kHz	4000000 Hz	20 nT	45 µT	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite pla...
RFM-09	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	1800000 Hz	0.1 µT	90 µT	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite pla...
RFM-10	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	1800000 Hz	1 nT	10 µT	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite pla...
RFM-11	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	1800000 Hz	1 nT	300 µT	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite pla...
RFM-12	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	1800000 Hz	1 nT	45 µT	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite pla...
RFM-13	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	1800000 Hz	1.5 µT	10 µT	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator;

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
											Square Helmholtz coils; Ferrite pla...
RFM-14	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	1800000 Hz	1.5 μ T	300 μ T	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite pla...
RFM-15	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	1800000 Hz	1.5 μ T	45 μ T	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite pla...
RFM-16	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	3600000 Hz	1 nT	45 μ T	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite pla...
RFM-17	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	3600000 Hz	1.5 μ T	45 μ T	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite pla...
RFM-18	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	4000000 Hz	20 nT	45 μ T	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite pla...
RFM-19	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	7200000 Hz	1 nT	45 μ T	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite pla...
RFM-20	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	7200000 Hz	1.5 μ T	45 μ T	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite pla...
RFM-21	P29	1	RF-MF + SMF (MF/HF-EMF + SMF)	Not reported	900000 Hz	0.1 μ T	45 μ T	E N/A N/A; PD N/A N/A		HT-1080(1)	ICOM IC-7300 HF/50 MHz base transceiver; Hewlett Packard 33120A function generator; Square Helmholtz coils; Ferrite pla...

Appendix A.3. Intermediate-frequency alternating electric fields

Appendix Table A.3. Exposure lookup for Intermediate-frequency alternating electric fields.

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
AEF-01	P17	2	TTFields (AEF)	Sine	100000 Hz	N/A N/A	N/A N/A	E 2.1 V/cm; PD N/A N/A		HCCC-9810(1), RBE(1)	TTFields delivery system (Healthy Life Innovation Medical Technology, Jiangsu, China)
AEF-02	P17	2	TTFields (AEF)	Sine	150000 Hz	N/A N/A	N/A N/A	E 1.3 V/cm; PD N/A N/A		HCCC-9810(1), RBE(1)	TTFields delivery system (Healthy Life Innovation Medical Technology, Jiangsu, China)
AEF-03	P17	2	TTFields (AEF)	Sine	150000 Hz	N/A N/A	N/A N/A	E 1.7 V/cm; PD N/A N/A		HCCC-9810(1), RBE(1)	TTFields delivery system (Healthy Life Innovation Medical

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
											Technology, Jiangsu, China)
AEF-04	P17	2	TTFields (AEF)	Sine	150000 Hz	N/A N/A	N/A N/A	E 2.1 V/cm; PD N/A N/A		HCCC-9810(1), RBE(1)	TTFields delivery system (Healthy Life Innovation Medical Technology, Jiangsu, China)
AEF-05	P17	2	TTFields (AEF)	Sine	200000 Hz	N/A N/A	N/A N/A	E 2.1 V/cm; PD N/A N/A		HCCC-9810(1), RBE(1)	TTFields delivery system (Healthy Life Innovation Medical Technology, Jiangsu, China)
AEF-06	P26	6	TTFields (AEF)	Sine (FM)	150000 Hz	N/A N/A	N/A N/A	E 1.5 V/cm; PD N/A N/A		MDA-MB-231(3), MCF-12(3)	Custom device: Tektronix AFG1062 function generator connected to a 4-channel relay module, Arduino Uno Rev3 SMD control...
AEF-07	P26	6	TTFields (AEF)	Sine (UM)	150000 Hz	N/A N/A	N/A N/A	E 1.5 V/cm; PD N/A N/A		MDA-MB-231(3), MCF-12(3)	Custom device: Tektronix AFG1062 function generator connected to a 4-channel relay module, Arduino Uno Rev3 SMD control...
AEF-08	P27	3	Tumor Treating Induced Fields (TTIF)	Sine	200000 Hz	N/A N/A	N/A N/A	E Not reported Not reported; PD N/A N/A		U87(3)	TTIF device: motor, wires, capacitor, and magnetic ring forming an LC resonance circuit (transformer-like electric fiel...
AEF-09	P34	5	TTFields (AEF)	Sine	150000 Hz	N/A N/A	N/A N/A	E 1.5 V/cm; PD N/A N/A		MDA-MB-231(3), HCC38(1), MCF-12(1)	In-house built TTField delivery device (Tektronix AFG1062 function generator, Arduino Uno Rev3 control board, 4-channel...
AEF-10	P34	3	TTFields (AEF)	Sine	150000 Hz	N/A N/A	N/A N/A	E 3 V/cm; PD N/A N/A		MDA-MB-231(1), HCC38(1), MCF-12(1)	In-house built TTField delivery device (Tektronix AFG1062 function generator, Arduino Uno Rev3 control board, 4-channel...
AEF-11	P34	3	TTFields (AEF)	Sine	150000 Hz	N/A N/A	N/A N/A	E 3.75 V/cm; PD N/A N/A		MDA-MB-231(1), HCC38(1), MCF-12(1)	In-house built TTField delivery device (Tektronix AFG1062 function generator, Arduino Uno Rev3 control board, 4-channel...
AEF-12	P34	3	TTFields (AEF)	Sine	150000 Hz	N/A N/A	N/A N/A	E 4.4 V/cm; PD N/A N/A		MDA-MB-231(1), HCC38(1), MCF-12(1)	In-house built TTField delivery device (Tektronix AFG1062 function generator, Arduino Uno Rev3 control board, 4-channel...
AEF-13	P34	3	TTFields (AEF)	Sine	150000 Hz	N/A N/A	N/A N/A	E 6 V/cm; PD N/A N/A		MDA-MB-231(1), HCC38(1), MCF-12(1)	In-house built TTField delivery device (Tektronix AFG1062 function generator, Arduino Uno Rev3 control board, 4-channel...
AEF-14	P39	1	AEF	Not reported	Not reported Not reported	N/A N/A	N/A N/A	E 0.26 V/cm; PD N/A N/A		B16F10(1)	Rotary triboelectric nanogenerator (TENG) based on 80 micrometer-thick PTFE film and copper electrodes on 80 mm diamete...
AEF-15	P49	1	TTFields (AEF)	Sine	100000 Hz	N/A N/A	N/A N/A	E 2.2 V/cm; PD N/A N/A		U251(1)	TEFT system (CL-301A, Antai

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
											Kangcheng Biotechnology Co., Ltd.)
AEF-16	P49	1	TTFields (AEF)	Sine	200000 Hz	N/A N/A	N/A N/A	E 1 V/cm; PD N/A N/A		U251(1)	TEFT system (CL-301A, Antai Kangcheng Biotechnology Co., Ltd.)
AEF-17	P49	1	TTFields (AEF)	Sine	200000 Hz	N/A N/A	N/A N/A	E 1.5 V/cm; PD N/A N/A		U251(1)	TEFT system (CL-301A, Antai Kangcheng Biotechnology Co., Ltd.)
AEF-18	P49	5	TTFields (AEF)	Sine	200000 Hz	N/A N/A	N/A N/A	E 2.2 V/cm; PD N/A N/A		U251(5)	TEFT system (CL-301A, Antai Kangcheng Biotechnology Co., Ltd.)
AEF-19	P49	1	TTFields (AEF)	Sine	300000 Hz	N/A N/A	N/A N/A	E 2.2 V/cm; PD N/A N/A		U251(1)	TEFT system (CL-301A, Antai Kangcheng Biotechnology Co., Ltd.)

Appendix A.4. Nanosecond pulsed electric fields

Appendix Table A.4. Exposure lookup for Nanosecond pulsed electric fields.

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
NPEF-01	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 100 kV/cm; PD N/A N/A	5 ns; 100 Hz; 10000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory-developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-02	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 100 kV/cm; PD N/A N/A	5 ns; 100 Hz; 2000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory-developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-03	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 100 kV/cm; PD N/A N/A	5 ns; 100 Hz; 4000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory-developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-04	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 100 kV/cm; PD N/A N/A	5 ns; 100 Hz; 6000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory-developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-05	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 100 kV/cm; PD N/A N/A	5 ns; 100 Hz; 8000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory-developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-06	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 70 kV/cm; PD N/A N/A	5 ns; 100 Hz; 10000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory-developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
NPEF-07	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 70 kV/cm; PD N/A N/A	5 ns; 100 Hz; 2000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory- developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-08	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 70 kV/cm; PD N/A N/A	5 ns; 100 Hz; 4000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory- developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-09	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 70 kV/cm; PD N/A N/A	5 ns; 100 Hz; 6000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory- developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-10	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 70 kV/cm; PD N/A N/A	5 ns; 100 Hz; 8000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory- developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-11	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 80 kV/cm; PD N/A N/A	5 ns; 100 Hz; 10000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory- developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-12	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 80 kV/cm; PD N/A N/A	5 ns; 100 Hz; 2000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory- developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-13	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 80 kV/cm; PD N/A N/A	5 ns; 100 Hz; 4000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory- developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-14	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 80 kV/cm; PD N/A N/A	5 ns; 100 Hz; 6000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory- developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-15	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 80 kV/cm; PD N/A N/A	5 ns; 100 Hz; 8000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory- developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-16	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 80 kV/cm; PD N/A N/A	5 ns; 1000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory- developed), 3-5 kV output, with custom

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
											microelectrodes (0.5 mm spac...
NPEF-17	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 90 kV/cm; PD N/A N/A	5 ns; 100 Hz; 10000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory-developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-18	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 90 kV/cm; PD N/A N/A	5 ns; 100 Hz; 2000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory-developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-19	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 90 kV/cm; PD N/A N/A	5 ns; 100 Hz; 4000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory-developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-20	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 90 kV/cm; PD N/A N/A	5 ns; 100 Hz; 6000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory-developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-21	P42	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 90 kV/cm; PD N/A N/A	5 ns; 100 Hz; 8000 pulses	MDA-MB-231(1)	Custom-built nanosecond pulse generator (laboratory-developed), 3-5 kV output, with custom microelectrodes (0.5 mm spac...
NPEF-22	P43	1	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 10 kV/cm; PD N/A N/A	300 ns; 2 Hz; 800 pulses	Hep3B(1)	Pulse generator device with two electrode needles, provided by Hangzhou RuiDi Biological Technology Co., LTD (China). W...
NPEF-23	P45	2	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 10 kV/cm; PD N/A N/A	50 ns; 2000 Hz	MCF-7(1), MCF-10A(1)	3D gold microelectrode device integrated with confocal microscope, pulse generator AVL-5-B-TR (AVTECH)
NPEF-24	P45	2	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 15 kV/cm; PD N/A N/A	50 ns; 2000 Hz	MCF-7(1), MCF-10A(1)	3D gold microelectrode device integrated with confocal microscope, pulse generator AVL-5-B-TR (AVTECH)
NPEF-25	P45	2	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 20 kV/cm; PD N/A N/A	50 ns; 2000 Hz	MCF-7(1), MCF-10A(1)	3D gold microelectrode device integrated with confocal microscope, pulse generator AVL-5-B-TR (AVTECH)
NPEF-26	P46	3	nsPEF	Not reported	N/A N/A	N/A N/A	N/A N/A	E 40 kV/cm; PD N/A N/A	100 ns; 200000 Hz; 60 pulses	U-87 MG(1), C6(1), GL261(1)	Custom-made pulse generator based on dual Marx circuit topology with 1.7 kV MOSFET switches and custom PCB electrodes w...

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
NPEF-27	P47	6	nsPEF	Square	N/A N/A	N/A N/A	N/A N/A	E 8 kV/cm; PD N/A N/A	100 ns; 10000 Hz; 100 pulses	A375(2), C32(1), MeWo(1), Colo- 829(1), Me45(1)	Square wave electroporator (100 ns-1 ms) developed at the Institute of High Magnetic Fields, VGTU, Vilnius, Lithuania

Appendix A.5. Ultrashort pulsed magnetic fields

Appendix Table A.5. Exposure lookup for Ultrashort pulsed magnetic fields.

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
PMF-01	P22	1	nsPMF	Square	N/A N/A	0.24 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 1 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-02	P22	1	nsPMF	Square	N/A N/A	0.24 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 10 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-03	P22	1	nsPMF	Square	N/A N/A	0.24 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 20 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-04	P22	1	nsPMF	Square	N/A N/A	0.24 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 5 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-05	P22	1	nsPMF	Square	N/A N/A	0.353 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 1 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-06	P22	1	nsPMF	Square	N/A N/A	0.353 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 10 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-07	P22	1	nsPMF	Square	N/A N/A	0.353 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 20 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-08	P22	1	nsPMF	Square	N/A N/A	0.353 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 5 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-09	P22	1	nsPMF	Square	N/A N/A	0.598 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 1 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
PMF-10	P22	1	nsPMF	Square	N/A N/A	0.598 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 10 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-11	P22	1	nsPMF	Square	N/A N/A	0.598 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 20 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-12	P22	1	nsPMF	Square	N/A N/A	0.598 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 5 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-13	P22	1	nsPMF	Square	N/A N/A	0.787 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 1 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-14	P22	1	nsPMF	Square	N/A N/A	0.787 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 10 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-15	P22	1	nsPMF	Square	N/A N/A	0.787 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 20 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-16	P22	1	nsPMF	Square	N/A N/A	0.787 mT	N/A N/A	E N/A N/A; PD N/A N/A	600 ns; 5 Hz	C6(1)	Transmission-line type square wave electromagnetic field generation device with mercury discharge switch and coaxial ca...
PMF-17	P22	1	μsPMF	Exponential	N/A N/A	151 mT	N/A N/A	E N/A N/A; PD N/A N/A	205 μs	C6(1)	Solenoid-based pulsed electromagnetic field generation device with square wave or exponential wave current source
PMF-18	P22	1	μsPMF	Exponential	N/A N/A	16.4 mT	N/A N/A	E N/A N/A; PD N/A N/A	205 μs	C6(1)	Solenoid-based pulsed electromagnetic field generation device with square wave or exponential wave current source
PMF-19	P22	1	μsPMF	Exponential	N/A N/A	311 mT	N/A N/A	E N/A N/A; PD N/A N/A	205 μs	C6(1)	Solenoid-based pulsed electromagnetic field generation device with square wave or exponential wave current source
PMF-20	P22	1	μsPMF	Exponential	N/A N/A	35.9 mT	N/A N/A	E N/A N/A; PD N/A N/A	205 μs	C6(1)	Solenoid-based pulsed electromagnetic field generation device with square wave or exponential wave current source
PMF-21	P22	1	μsPMF	Square	N/A N/A	122 mT	N/A N/A	E N/A N/A; PD N/A N/A	100 μs	C6(1)	Solenoid-based pulsed

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
											electromagnetic field generation device with square wave or exponential wave current source
PMF-22	P22	1	µsPMF	Square	N/A N/A	363 mT	N/A N/A	E N/A N/A; PD N/A N/A	100 µs	C6(1)	Solenoid-based pulsed electromagnetic field generation device with square wave or exponential wave current source
PMF-23	P22	1	µsPMF	Square	N/A N/A	45.4 mT	N/A N/A	E N/A N/A; PD N/A N/A	100 µs	C6(1)	Solenoid-based pulsed electromagnetic field generation device with square wave or exponential wave current source
PMF-24	P22	1	µsPMF	Square	N/A N/A	854 mT	N/A N/A	E N/A N/A; PD N/A N/A	100 µs	C6(1)	Solenoid-based pulsed electromagnetic field generation device with square wave or exponential wave current source

Appendix A.6. RF/mobile-frequency electromagnetic exposures

Appendix Table A.6. Exposure lookup for RF/mobile-frequency electromagnetic exposures.

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
RF-01	P30	3	REAC (RF-EMF)	Not reported	Not reported Not reported	Not reported Not reported	N/A N/A	E N/A N/A; PD N/A N/A		MCF-7(3)	BENE 110 (ASMED, Italy); Radioelectric asymmetric conveyance
RF-02	P31	6	LTE (UHF-EMF)	LTE (WCDMA modulation)	1700000000 Hz	N/A N/A	N/A N/A	E N/A N/A; PD N/A N/A; SAR 1		ASC(1), Huh7(1), Hep3B(1), HeLa(1), SH-SY5Y(1), IMR-90(1)	Radial transmission line (RTL) exposure system with conical antenna, installed in incubator with forced refrigerated wa...
RF-03	P31	6	LTE (UHF-EMF)	LTE (WCDMA modulation)	1700000000 Hz	N/A N/A	N/A N/A	E N/A N/A; PD N/A N/A; SAR 2		ASC(1), Huh7(1), Hep3B(1), HeLa(1), SH-SY5Y(1), IMR-90(1)	Radial transmission line (RTL) exposure system with conical antenna, installed in incubator with forced refrigerated wa...
RF-04	P35	5	LTE (UHF-EMF)	LTE (WCDMA modulation)	1760000000 Hz	N/A N/A	N/A N/A	E N/A N/A; PD N/A N/A; SAR 0.4		Huh7(2), HeLa(1), ASC(1), B103(1)	Radial transmission line (RTL) exposure system with E4438C signal generator (Keysight Technologies), customized power a...
RF-05	P35	3	LTE (UHF-EMF)	LTE (WCDMA modulation)	1760000000 Hz	N/A N/A	N/A N/A	E N/A N/A; PD N/A N/A; SAR 1		Huh7(1), HeLa(1), B103(1)	Radial transmission line (RTL) exposure system with E4438C signal generator (Keysight Technologies), customized power a...
RF-06	P35	5	LTE (UHF-EMF)	LTE (WCDMA modulation)	1760000000 Hz	N/A N/A	N/A N/A	E N/A N/A; PD N/A N/A; SAR 4		Huh7(2), HeLa(1), ASC(1), B103(1)	Radial transmission line (RTL) exposure system with E4438C signal generator (Keysight Technologies), customized power a...
RF-07	P35	4	LTE (UHF-EMF)	LTE (WCDMA modulation)	1760000000 Hz	N/A N/A	N/A N/A	E N/A N/A; PD N/A N/A; SAR 8		Huh7(1), HeLa(1), ASC(1), B103(1)	Radial transmission line (RTL) exposure system with E4438C

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
											signal generator (Keysight Technologies), customized power a...
RF-08	P37	3	RF-EMF (MF-EMF)	Amplitude-modulated sine	13560000 Hz	N/A N/A	N/A N/A	E N/A N/A; PD N/A N/A		HT29(1), SW480(1), LoVo(1)	LabEHY-200 (Oncotherm Kft., Budapest, Hungary) with novel in vitro applicator
RF-09	P37	3	RF-EMF (MF-EMF)	Continuous sine	13560000 Hz	N/A N/A	N/A N/A	E N/A N/A; PD N/A N/A		HT29(1), SW480(1), LoVo(1)	LabEHY-200 (Oncotherm Kft., Budapest, Hungary) with novel in vitro applicator

Appendix A.7. Millimeter-wave/terahertz radiative exposures

Appendix Table A.7. Exposure lookup for Millimeter-wave/terahertz radiative exposures.

ID	Papers	Rows	Field label	Wave	Frequency	AC mag. intensity	DC/static	E / power / SAR	Pulse structure	Cells	Device
EHF-01	P11	6	MMW (EHF-EMF)	Sine regime (frequency skipping across ...	75000000000-1100000000000 Hz	N/A N/A	N/A N/A	E N/A N/A; PD 0.2 mW/cm ²		H1299(6)	Scanning 8360B Series Swept Signal Generator with 8757D Scalar Network Analyzer (Agilent), solid-state frequency multip...
EHF-02	P12	22	MMW (EHF-EMF)	Sine regime (frequency skipping across ...	750000000000-1050000000000 Hz	N/A N/A	N/A N/A	E N/A N/A; PD 0.2 mW/cm ²		MCF-10A(15), H1299(7)	Scanning 8360B Series Swept Signal Generator with 8757D Scalar Network Analyzer (Agilent), solid-state frequency multip...
EHF-03	P32	4	MMW (EHF-EMF)	Continuous sine	352000000000 Hz	N/A N/A	N/A N/A	E N/A N/A; PD 0.16 mW/cm ²		A375(4)	KFA-100A MMW therapeutic instrument (Zhongcheng Kangfu Technology)
EHF-04	P33	8	THz radiation (EHF-EMF)	Continuous	1400000000000 Hz	N/A N/A	N/A N/A	E N/A N/A; PD 7.4 mW/cm ²		Ramos(2), Jurkat(2), MCF-7(2), PBMCs(1), LNCaP(1)	IMPATT (impact ionization avalanche transit-time) diode-based radiation source with rectangular horn antenna (14 mm x 8...